BME302 Unit V – Centrifugal & Reciprocating Pumps

Lecture Notes · Fluid Mechanics and Fluid Machines

Dr. Abhishek Pandey, Department of Mechanical Engineering, G. L. Bajaj Institute of Technology & Management

How to use these notes. Every topic, worked example and question carries one tag. CORE must-know syllabus material at AKTU exam level: master these first. HOTS higher-order thinking: analyse, compare, justify or design (Bloom K4 to K6). ADVANCED beyond the exam level for most students: shaded orange, optional reading.

Unit overview

Unit V applies the turbine ideas of Unit IV in reverse: a pump takes in shaft work and raises the energy of a liquid. Centrifugal pumps do this continuously through a rotating impeller; reciprocating pumps do it in batches with a piston. The unit maps to **CO5: Understand the functioning of centrifugal and reciprocating pumps (K2)** and builds on CO4 for the velocity-triangle analysis. It is planned for 8 lectures; notation follows R.K. Bansal and Som, Biswas & Chakraborty.

Lecture	Topic	Key outcome
L1	Centrifugal pump: working, main parts, priming, classification	Identify pump types and their parts
L2	Heads and efficiencies of a centrifugal pump	Define manometric head and the four efficiencies
L3	Vector diagrams, work done by impeller, effect of blade shape	Derive $\text{work} = V_{w2}u_2/g$
L4	Minimum starting speed, multistage pumps, specific speed, model laws	Size and compare pumps
L5	Cavitation and separation, NPSH, Thoma’s cavitation factor	Check a pump installation for cavitation
L6	Reciprocating pump theory, discharge, work, slip	Compute discharge, power and slip
L7	Indicator diagram, effect of acceleration and friction, separation	Find maximum speed without separation
L8	Air vessels; comparison of centrifugal and reciprocating pumps	Quantify the work saved by air vessels

Common notation: suffix 1 = impeller inlet (eye), 2 = impeller outlet. u = blade velocity, V = absolute velocity, V_r = relative velocity, V_w = whirl component, V_f = flow component, θ = vane angle at inlet, ϕ = vane angle at outlet, D = impeller diameter, B = impeller width, N = speed in rpm, Q = discharge, H_m = manometric head, η_{mano}, η_m, η_v, η_o = manometric, mechanical, volumetric and overall efficiency. For the reciprocating pump: A = piston area, L = stroke length, r = crank radius ($L = 2r$), ω = crank angular velocity, θ = crank angle, a = pipe area, h_s and h_d = suction and delivery heads, L_s and L_d = suction and delivery pipe lengths.

0. Before you start: basic ideas you need CORE

This section collects the basic ideas that the whole unit uses again and again. Read it first if pump problems feel hard to start. Every later section assumes you are at ease with velocity triangles, units of head and power, the energy equation and the motion of a piston.

> **IN SIMPLE WORDS**
>
> A pump is a machine that gives energy to water. A centrifugal pump spins a wheel that flings the water outward, like mud flying off a bicycle tyre. A reciprocating pump pushes water with a piston, like a bicycle pump. Engineers measure the energy given to the water as a height, called head, and the effort of the motor as power. You need four small ideas: velocity triangles on a spinning vane, units of head and power, the energy equation with losses, and the motion of a piston.

0.1 Velocity triangles and impeller geometry CORE

The **impeller** is a wheel with curved vanes. Water enters near the centre (the eye) and leaves at the rim. At any point on a vane there are three velocities. Each one is a vector, which means it has a size and a direction.

- **Blade velocity** u: the speed of the vane itself, along the tangent to the circle. It is $u = \omega r = \pi DN/60$, where D is the diameter in m and N is the speed in rpm.
- **Relative velocity** V_r: the velocity of the water as seen by someone riding on the vane. It points along the vane.
- **Absolute velocity** V: the velocity of the water as seen by someone standing still beside the pump. It is the vector sum of the other two: $V = u + V_r$ (add them head to tail).

Split the absolute velocity V into two parts at right angles. The **whirl component** V_w is along the blade velocity (the sideways, swirling part). The **flow component** V_f is along the radius (the part that carries the water through the pump). The flow component alone fixes the discharge, because $Q = \pi DBV_f$, where B is the impeller width. The outlet vane angle ϕ is the angle between V_r and the reversed blade velocity, so from the right-angled triangle $\tan\phi = V_f/(u - V_w)$.

Symbol	Meaning	SI unit	Typical value
D, B	Impeller diameter, impeller width	m	D from 0.1 to $0.6\,\mathrm{m}$; B a few cm
N	Speed	rpm	1450 or 2900 for a motor on 50 Hz supply
$u = \pi DN/60$	Blade velocity	m/s	10 to $40\,\mathrm{m/s}$
V_r	Relative velocity (along the vane)	m/s	A few m/s
V	Absolute velocity	m/s	5 to $30\,\mathrm{m/s}$
V_w, V_f	Whirl and flow components of V	m/s	V_f is 1 to $5\,\mathrm{m/s}$
θ, ϕ	Vane angles at inlet and outlet	degrees	ϕ from 15° to 35°

Example 0.1 CORE An impeller of diameter 300 mm runs at 1450 rpm. Find the blade velocity at the rim.

1. $D = 300\,\mathrm{mm} = 0.3\,\mathrm{m}$
2. $u = \dfrac{\pi DN}{60} = \dfrac{\pi \times 0.3 \times 1450}{60} = 22.78\,\mathrm{m/s}$

Example 0.2 CORE At the outlet of an impeller the blade velocity is $u_2 = 20\,\mathrm{m/s}$, the flow velocity is $V_{f2} = 4\,\mathrm{m/s}$ and the outlet vane angle is $\phi = 45^\circ$. Find V_{w2}, V_{r2} and V_2.

1. $V_{w2} = u_2 - \dfrac{V_{f2}}{\tan\phi} = 20 - \dfrac{4}{\tan 45^\circ} = 16\,\mathrm{m/s}$
2. $V_{r2} = \dfrac{V_{f2}}{\sin\phi} = \dfrac{4}{\sin 45^\circ} = 5.657\,\mathrm{m/s}$
3. $V_2 = \sqrt{V_{w2}^2 + V_{f2}^2} = \sqrt{16^2 + 4^2} = 16.49\,\mathrm{m/s}$
4. Check: $V_{r2}^2 = (u_2 - V_{w2})^2 + V_{f2}^2 = 4^2 + 4^2 = 32$, and $5.657^2 = 32$

Example 0.3 CORE The impeller of Example 0.1 is 20 mm wide at the rim, and the flow velocity there is 2 m/s. Find the discharge.

1. Flow area at the rim: $\pi DB = \pi \times 0.3 \times 0.02 = 0.01885\,\mathrm{m^2}$
2. $Q = \pi DBV_f = 0.01885 \times 2 = 0.0377\,\mathrm{m^3/s}$, that is $37.7\,\mathrm{L/s}$

0.2 Pressure, head and power units CORE

Pump data come in many units. Change them to SI units first. A pump gives energy to a liquid. **Head** is that energy per unit weight of liquid, in metres. **Water power** is the useful power the liquid receives, $\rho g Q H$.

Given in	Convert to	Rule	Example
mm	m	Divide by 1000	250 mm = 0.25 m
L/s	m^3/s	Divide by 1000	40 L/s = 0.04 m^3/s
m^3/h	m^3/s	Divide by 3600	36 m^3/h = 0.01 m^3/s
kPa, bar	m of water	$h = p/(\rho g)$	300 kPa → 30.58 m
rpm	rad/s	$\omega = 2\pi N/60$	1450 rpm → 151.8 rad/s
W	kW	Divide by 1000	1962 W = 1.962 kW
hp	kW	Multiply by 0.7457	1 hp = 0.7457 kW

Water power in kW is $\rho g Q H/1000$. Shaft power is larger, because $P = \text{water power}/\eta_o$.

Example 0.4 CORE A pressure gauge on a water line reads 300 kPa. Express this as a head of water.

1. $h = \dfrac{p}{\rho g} = \dfrac{300\,000}{1000 \times 9.81} = 30.58\ \text{m}$

Example 0.5 CORE A pump delivers 36 m^3/h of water against a head of 20 m. Find the water power in kW and in hp, and the shaft power if $\eta_o = 70\%$.

1. $Q = \dfrac{36}{3600} = 0.01\ m^3/s$
2. $\text{Water power} = \dfrac{\rho g Q H}{1000} = \dfrac{1000 \times 9.81 \times 0.01 \times 20}{1000} = 1.962\ \text{kW}$, which is $\dfrac{1.962}{0.7457} = 2.63\ \text{hp}$
3. $P = \dfrac{1.962}{0.70} = 2.80\ \text{kW}$

0.3 Energy equation with losses: a recap CORE

The energy equation of Unit III says that the total head of flowing liquid falls along a pipe by the head lost to friction and fittings. The total head is the sum of pressure head, velocity head and datum head. With a pump between sections 1 and 2, the pump adds the head H_m to the liquid:

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 + H_m = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L$$

Here h_L is the head lost to friction and fittings between the sections. Use gauge pressure at both points, or absolute pressure at both. The surface of an open tank has $p = 0$ gauge and, if the tank is large, $V \approx 0$.

Example 0.6 CORE A pump lifts water from a large open sump to a large open tank whose water surface is 12 m higher. The total friction loss in the pipes is 3 m. Find the head the pump must give.

1. Both surfaces are open, so $p_1 = p_2 = 0$ gauge, and $V_1 \approx V_2 \approx 0$.
2. $H_m = (z_2 - z_1) + h_L = 12 + 3 = 15\ \text{m}$

0.4 Reciprocating motion basics CORE

A reciprocating pump has a **piston** that moves to and fro inside a **cylinder**. A rotating **crank** of radius r drives it through a **connecting rod**. The piston moves from one end of the cylinder (inner dead centre) to the other end (outer dead centre). The distance between the two ends is the **stroke** $L = 2r$. One turn of the crank gives two strokes. The crank turns at the angular velocity $\omega = 2\pi N/60$.

Let the crank angle $\theta = \omega t$ be measured from the inner dead centre. If the connecting rod is long compared with the crank, the piston moves like the shadow of the crank pin on the cylinder axis. This is **simple harmonic motion**. Then the displacement, velocity and acceleration of the piston are:

$$x = r\,(1 - \cos\theta), \qquad v = \frac{dx}{dt} = r\omega \sin\theta, \qquad a = \frac{dv}{dt} = \omega^2 r \cos\theta$$

The velocity is largest at mid-stroke ($\theta = 90^\circ$), where $v_{\max} = \omega r$. The acceleration is largest at the two ends of the stroke ($\theta = 0^\circ$ and 180°), where $a_{\max} = \omega^2 r$, and it is zero at mid-stroke. In one revolution the piston travels $2L$, so its mean speed is $2LN/60 = LN/30$.

Example 0.7 CORE A crank of radius 0.15 m turns at 60 rpm. Find the maximum piston velocity, the maximum acceleration, the mean piston speed and the displacement at $\theta = 60^\circ$.

1. $\omega = \dfrac{2\pi \times 60}{60} = 6.283\ \text{rad/s}$, and $L = 2r = 0.3\ \text{m}$
2. $v_{\max} = \omega r = 6.283 \times 0.15 = \ 0.942\ \text{m/s}$
3. $a_{\max} = \omega^2 r = 6.283^2 \times 0.15 = \ 5.92\ \text{m/s}^2$
4. Mean speed $= \dfrac{LN}{30} = \dfrac{0.3 \times 60}{30} = \ 0.6\ \text{m/s}$, so $v_{\max}$ is $\pi/2 = 1.571$ times the mean speed
5. $x = r\,(1 - \cos 60^\circ) \ = 0.15 \times 0.5 = \ 0.075\ \text{m}$

Common mistakes and exam tips.

- Putting a diameter in mm straight into $u = \pi DN/60$. Change it to metres first.
- Using the absolute velocity V where the relative velocity V_r or the whirl component V_w is meant. Draw the triangle and label each side.
- Using rpm where rad/s is needed. Convert with $\omega = 2\pi N/60$.
- Forgetting that shaft power is larger than water power. Divide the water power by η_o.
- Mixing up the crank radius r and the stroke $L = 2r$ in reciprocating-pump formulas.

Check your understanding.

1. CORE Convert 45 L/s to m^3/s. *(It is $0.045\ \text{m}^3/\text{s}$.)*
2. CORE The stroke of a pump is 0.4 m. What is the crank radius? *(0.2 m, since $L = 2r$.)*
3. HOTS Where in the stroke is the piston acceleration zero, and where is the velocity zero? *(Acceleration is zero at mid-stroke. Velocity is zero at the two ends.)*

1. Centrifugal pump: working, parts and classification (L1)

CORE

A centrifugal pump is a rotodynamic machine that raises the pressure of a liquid by whirling it in a rotating impeller. It works on the **forced vortex principle**: when a liquid is rotated at angular velocity ω, the pressure head rises with the square of the radius.

$$\text{Rise in pressure head} = \frac{\omega^2 r_2^2}{2g} - \frac{\omega^2 r_1^2}{2g} = \frac{u_2^2 - u_1^2}{2g}$$

IN SIMPLE WORDS

A centrifugal pump works like a spinning merry-go-round for water. The vanes throw the water outward, just as you feel pushed outward on a fast ride. The faster the wheel turns and the larger its radius, the more pressure the water gains. In the formula, $u = \omega r$ is the speed of a point on the wheel, and dividing u^2 by $2g$ turns it into a height in metres of water.

Liquid enters the eye of the impeller axially, is flung outward by the vanes, and leaves at the periphery with high velocity and pressure. The casing then converts much of the leaving kinetic energy into further pressure head.

Derivation from scratch: pressure rise in a rotating impeller (forced vortex)

Aim. Prove that the pressure head at radius r_2 exceeds the pressure head at radius r_1 by $\left(u_2^2 - u_1^2 \right) / \left(2g \right)$ when the liquid turns with the impeller.

Assumptions.

1. The liquid turns with the impeller as a solid body at constant angular velocity ω, with no flow relative to the vanes. This is the situation when the delivery valve is closed.
2. The liquid is incompressible, with constant density ρ.
3. The height difference between the two radii is negligible.

Step 1. Choose an element. Take a thin slice of liquid at radius r from the axis. Its thickness along the radius is dr and its area, at right angles to the radius, is dA. Its volume is $dA\,dr$ and its mass is $\rho\,dA\,dr$.

Step 2. Acceleration of the element. The element moves on a circle of radius r at speed ωr. It therefore has an acceleration towards the axis (centripetal acceleration):

$$a = \frac{\left(\omega r \right)^2}{r} = \omega^2 r$$

By Newton's second law the net force towards the axis must be

$$F = \text{mass} \times a = \rho\,dA\,dr\,\omega^2 r$$

Step 3. Forces on the element. Only the pressure on the two faces acts along the radius. The inner face is at pressure p and is pushed outward (away from the axis) by the force $p\,dA$. The outer face is at pressure $p + dp$ and is pushed inward (towards the

axis) by the force $(p + dp)\,dA$. The net force towards the axis is

$$(p + dp)\,dA - p\,dA = dp\,dA$$

Step 4. Apply Newton's second law. Equate the net inward force to the force needed in Step 2, and cancel the common factor dA:

$$dp\,dA = \rho\,dA\,dr\,\omega^2 r \quad \Rightarrow \quad dp = \rho\,\omega^2 r\,dr$$

Step 5. Integrate from r_1 to r_2. The pressure rises from p_1 to p_2 as the radius grows from r_1 to r_2:

$$\int_{p_1}^{p_2} dp = \rho\,\omega^2 \int_{r_1}^{r_2} r\,dr \quad \Rightarrow \quad p_2 - p_1 = \rho\,\omega^2 \left[\frac{r^2}{2}\right]_{r_1}^{r_2} = \frac{\rho\,\omega^2\,(r_2^2 - r_1^2)}{2}$$

Step 6. Convert to head. Divide by ρg and use $u = \omega r$:

$$\frac{p_2 - p_1}{\rho g} = \frac{\omega^2 r_2^2}{2g} - \frac{\omega^2 r_1^2}{2g} = \frac{u_2^2 - u_1^2}{2g}$$

Result.

$$\text{Rise in pressure head} = \frac{\omega^2 r_2^2}{2g} - \frac{\omega^2 r_1^2}{2g} = \frac{u_2^2 - u_1^2}{2g}$$

What it tells us. The pressure head grows with the square of the speed and with the square of the radius, so a larger or faster impeller makes more head. Doubling the speed gives four times the head. This is the head the pump can produce before any flow starts, so it fixes the minimum starting speed in 4.1. When liquid flows through the vanes, the fall in relative velocity adds a further pressure rise, which the casing and Euler's equation in 3.2 take into account.

1.1 Main parts CORE

IN SIMPLE WORDS

A centrifugal pump has only a few parts, and each has one job. The impeller is the spinning wheel that gives energy to the water. The casing is the shell that collects the fast water and turns its speed into pressure. The suction pipe brings water in, and the delivery pipe takes it out. The foot valve is a one-way gate at the bottom of the suction pipe, like the end of a straw held shut with a finger, so the pipe stays full when the pump stops.

1. **Impeller** — the rotating wheel with a set of backward-curved vanes, mounted on a shaft driven by an electric motor. It is the only part that adds energy to the liquid.
2. **Casing** — an air-tight passage around the impeller. It collects the liquid and converts kinetic energy into pressure energy. Three common designs are listed in 1.3.
3. **Suction pipe with foot valve and strainer** — connects the sump to the eye of the impeller. The foot valve is a non-return valve that keeps the pipe full when the pump stops; the strainer keeps out solids.

4. **Delivery pipe with delivery valve** — carries the liquid to the required height. The delivery valve regulates flow and is kept closed at starting.

1.2 Priming CORE

> **IN SIMPLE WORDS**
>
> Priming means filling the pump with water before you start it. Think of sucking a drink through a straw: it works only if the straw is not full of air. Air is about 800 times lighter than water, so an impeller spinning in air makes almost no pressure. It cannot pull water up the suction pipe. So the pump spins but delivers nothing until it is primed.

Priming means filling the suction pipe, casing and delivery pipe up to the delivery valve with liquid so that no air remains. It is essential because the head developed depends on the density of the fluid: an impeller running in air develops the same head in metres of air, which is only about $1/800$ of the same head in water. That pressure is far too small to lift water up the suction pipe, so an unprimed pump runs without delivering. Self-priming pumps and pumps placed below the sump level (flooded suction) avoid this step.

1.3 Classification of centrifugal pumps CORE

Basis	Types	Remarks
Casing	Volute casing · Vortex casing · Casing with guide blades (diffuser, turbine pump)	Volute: area grows along the spiral, eddy losses high. Vortex: circular whirlpool chamber between impeller and volute reduces eddy loss. Guide blades: highest efficiency, used in multistage pumps
Impeller	Closed (shrouded) · Semi-open · Open	Closed for clean water; semi-open and open for sewage, slurry and liquids with solids
Direction of flow	Radial · Mixed · Axial	Radial for high head and small discharge; axial (propeller pump) for low head and large discharge
Entry of liquid	Single suction · Double suction	Double suction halves the eye velocity and balances axial thrust
Number of stages	Single-stage · Multistage	Impellers in series raise head; impellers in parallel raise discharge
Head	Low (below 15 m) · Medium (15–40 m) · High (above 40 m)	Indicative ranges
Specific speed	Slow, medium and fast radial flow · Mixed flow · Axial flow	See 4.3

1.4 Working sequence CORE

1. Prime the pump, close the delivery valve and start the motor.
2. The impeller sets up a forced vortex; the pressure rise at the outlet lifts the liquid through the suction pipe into the eye.
3. Once the impeller reaches its rated speed, open the delivery valve gradually.
4. Liquid leaving the impeller passes through the casing, where velocity falls and pressure rises, and flows up the delivery pipe.
5. To stop, close the delivery valve first, then switch off the motor, so that the column of water does not run back through the pump.

Common mistakes and exam tips.

- Writing the head rise with the diameter in place of the radius. Either use $r = D/2$ in $\omega^2 r^2/2g$, or use $u = \pi DN/60$ directly.
- Forgetting to prime the pump. A pump that spins but gives no water is usually unprimed, not faulty.
- Mixing up the two ways of adding pumps. Impellers in series add head. Pumps in parallel add discharge.
- Starting or stopping in the wrong order. Start with the delivery valve closed and open it slowly. Stop by closing it first, then switching off the motor.
- In a 10-mark answer, draw and label a sketch of the pump with impeller, casing, suction pipe, foot valve, strainer and delivery valve.

Check your understanding.

1. CORE In what order do you close the valve and switch off the motor when stopping? *(Close the delivery valve first, then switch off the motor.)*
2. CORE What does the foot valve do? *(It keeps the suction pipe full when the pump stops.)*
3. HOTS The speed of an impeller is doubled. By what factor does the pressure-head rise change? *(It becomes four times as large, since $u \propto N$ and the head rise is proportional to u^2.)*

2. Heads and efficiencies of a centrifugal pump (L2) CORE

The useful output of a pump is the head it gives the liquid, measured between its inlet and outlet flanges. The input is shaft power. The ratios between the stages in between define four efficiencies.

IN SIMPLE WORDS

A pump is paid for by how high and how much water it moves, and what it costs in electricity. The head is the height the pump must push the water, plus the extra height that pipe friction takes away. The motor power is always larger than the power the water finally gets. Some power is lost in eddies, some in bearings, and some water leaks back. Each efficiency is simply output divided by input at one stage.

2.1 Heads in a pump installation CORE

IN SIMPLE WORDS

Think of carrying water in a bucket up a hill. The plain climb is the static head, the sum of the suction lift h_s and the delivery height h_d. The rough path costs extra effort, which is the friction head in the pipes. The water also leaves the pipe with some speed, and that speed energy is spent too. Add all of these and you get the manometric head H_m, the number a pair of pressure gauges would show across the pump.

Head	Symbol	Meaning
Suction head	h_s	Vertical height of the pump centre line above the sump water level
Delivery head	h_d	Vertical height of the delivery outlet (or tank water level) above the pump centre line
Static head	H_{st}	$H_{\text{st}} = h_s + h_d$, the net vertical lift of the liquid
Friction heads	h_{fs}, h_{fd}	Losses in the suction and delivery pipes
Manometric head	H_m	Head against which the pump must work, read by gauges at the pump inlet and outlet

The manometric head can be written three equivalent ways:

$$H_m = h_s + h_d + h_{fs} + h_{fd} + \frac{V_d^2}{2g}$$

$$H_m = \left(\frac{p_o}{\rho g} + \frac{V_o^2}{2g} + Z_o\right) - \left(\frac{p_i}{\rho g} + \frac{V_i^2}{2g} + Z_i\right)$$

$$H_m = \frac{V_{w2} u_2}{g} - (\text{losses in impeller and casing})$$

Here o and i denote the pump outlet and inlet flanges, and V_d is the velocity in the delivery pipe. In most problems $V_d^2/(2g)$ is small and is either given or neglected; the first form is the one used for an installation, the second for a test, and the third links H_m to the impeller.

Derivation from scratch: manometric head of an installation

Aim. Prove that $H_m = h_s + h_d + h_{fs} + h_{fd} + V_d^2/\,(\,2g\,)$, and show how the other two forms follow.

Assumptions.

1. Steady flow of an incompressible liquid.
2. The sump and the delivery tank are large and open to the atmosphere, so the sump surface has $V \approx 0$.
3. The inlet and outlet flanges i and o are at the pump centre line. Take that line as the datum, so $Z_i = Z_o = 0$. The sump surface is then at height $-h_s$ and the delivery outlet is at height $+h_d$.
4. h_{fs} and h_{fd} are the friction (and fitting) losses in the suction and delivery pipes. V_d is the velocity of the liquid leaving the delivery pipe.

Step 1. Energy equation on the suction side. Write the energy equation from the sump surface (pressure p_{atm}, velocity zero, height $-h_s$) to the pump inlet flange i. The liquid loses h_{fs} on the way:

$$\frac{p_{\text{atm}}}{\rho g} + 0 - h_s = \frac{p_i}{\rho g} + \frac{V_i^2}{2g} + 0 + h_{fs}$$

Step 2. Total head at the pump inlet. Rearrange Step 1:

$$\frac{p_i}{\rho g} + \frac{V_i^2}{2g} + Z_i = \frac{p_{\text{atm}}}{\rho g} - h_s - h_{fs}$$

Step 3. Energy equation on the delivery side. Write the energy equation from the pump outlet flange o to the end of the delivery pipe (pressure p_{atm}, velocity V_d, height $+h_d$). The liquid loses h_{fd} on the way. The velocity energy $V_d^2/2g$ with which it leaves is not recovered:

$$\frac{p_o}{\rho g} + \frac{V_o^2}{2g} + Z_o = \frac{p_{\text{atm}}}{\rho g} + h_d + h_{fd} + \frac{V_d^2}{2g}$$

Step 4. Definition of manometric head. The manometric head is the total head at the outlet flange minus the total head at the inlet flange. This is the second form in the notes, the one used when gauges are read:

$$H_m = \left(\frac{p_o}{\rho g} + \frac{V_o^2}{2g} + Z_o\right) - \left(\frac{p_i}{\rho g} + \frac{V_i^2}{2g} + Z_i\right)$$

Step 5. Subtract. Put the right-hand sides of Steps 3 and 2 into Step 4. The terms $p_{\text{atm}}/\rho g$ cancel:

$$H_m = \left(\frac{p_{\text{atm}}}{\rho g} + h_d + h_{fd} + \frac{V_d^2}{2g}\right) - \left(\frac{p_{\text{atm}}}{\rho g} - h_s - h_{fs}\right) = h_s + h_d + h_{fs} + h_{fd} + \frac{V_d^2}{2g}$$

Step 6. Link to the impeller. Section 3.2 shows that the impeller gives each newton of liquid the head $V_{w2}u_2/g$. Part of this head, h_L, is lost in the impeller and casing as eddies and friction. The rest is the head gained between the flanges:

$$H_m = \frac{V_{w2}u_2}{g} - h_L$$

Result.

$$H_m = h_s + h_d + h_{fs} + h_{fd} + \frac{V_d^2}{2g}$$

What it tells us. The pump must supply the static lift, the friction in both pipes and the velocity head of the delivery pipe. Longer or narrower pipes raise h_{fs} and h_{fd}, so the pump must work against a higher H_m. The impeller head $V_{w2}u_2/g$ is always larger than H_m, because of the hydraulic losses h_L.

2.2 Efficiencies CORE

IN SIMPLE WORDS

Between the motor and the water there are three places where power is lost. In the bearings and glands some power turns into heat. In the impeller and casing, eddies and friction waste some of the head. At the wearing rings some water leaks back to the eye. Each efficiency compares output with input at one of these stages. The overall efficiency η_o is water power divided by shaft power, and it equals the manometric efficiency times the mechanical efficiency.

Efficiency	Definition	Expression
Manometric η_{mano}	Manometric head / head imparted by impeller	$\eta_{\text{mano}} = gH_m/\left(V_{w2}u_2\right)$
Mechanical η_m	Power delivered by impeller to liquid / shaft power	$\eta_m = \rho Q\left(V_{w2}u_2\right)/P$
Volumetric η_v	Liquid actually delivered / liquid passing through impeller	$\eta_v = Q/\left(Q+q\right)$, q = leakage
Overall η_o	Water power / shaft power	$\eta_o = \rho gQH_m/P = \eta_{\text{mano}} \times \eta_m$

$$P = \frac{\rho g Q H_m}{1000\,\eta_o}\ \text{kW}, \qquad \eta_o = \eta_{\text{mano}} \times \eta_m$$

Typical values: η_{mano} 0.75–0.85, η_m 0.95–0.98, η_o 0.70–0.85 for well-designed water pumps.

Losses that lower these efficiencies: (a) hydraulic losses — shock at the impeller entry, friction in vane passages and eddies in the casing, which reduce η_{mano}; (b) mechanical losses — friction in bearings and glands and disc friction on the impeller shrouds, which reduce η_m; (c) leakage past the wearing rings from the outlet back to the eye, which reduces η_v.

Derivation from scratch: overall efficiency and shaft power

Aim. Prove that $\eta_o = \eta_{\text{mano}} \times \eta_m$ and that $P = \rho g Q H_m / \left(1000\,\eta_o\right)$ kW.

Assumptions.

1. The discharge Q is the discharge actually delivered by the pump.
2. H_m is the manometric head and $V_{w2}u_2/g$ is the head given by the impeller (Section 3.2).

Step 1. Write the three powers. The shaft power P goes in. The impeller passes the power $\rho Q V_{w2} u_2$ to the liquid. The water power, $\rho g Q H_m$, is the useful output.

Step 2. Write the two stage efficiencies from the table.

$$\eta_{\text{mano}} = \frac{H_m}{V_{w2}u_2/g} = \frac{gH_m}{V_{w2}u_2}, \qquad \eta_m = \frac{\rho Q V_{w2} u_2}{P}$$

Step 3. Multiply them. The factor $V_{w2}u_2$ appears once above and once below, and cancels:

$$\eta_{\text{mano}} \times \eta_m = \frac{gH_m}{V_{w2}u_2} \times \frac{\rho Q V_{w2} u_2}{P} = \frac{\rho g Q H_m}{P}$$

Step 4. Recognise the overall efficiency. The right-hand side is water power divided by shaft power, which is the definition of η_o:

$$\eta_o = \frac{\rho g Q H_m}{P} = \eta_{\text{mano}} \times \eta_m$$

Step 5. Solve for the shaft power. Rearrange, and divide by 1000 to change W into kW:

$$P = \frac{\rho g Q H_m}{\eta_o}\ \text{W} = \frac{\rho g Q H_m}{1000\,\eta_o}\ \text{kW}$$

Result.

$$\eta_o = \eta_{\text{mano}} \times \eta_m, \qquad P = \frac{\rho g Q H_m}{1000\,\eta_o}\ \text{kW}$$

What it tells us. The overall efficiency can never exceed either stage efficiency, so every loss lowers it. The shaft power is the water power divided by η_o, which is why the motor must be larger than the water power alone suggests.

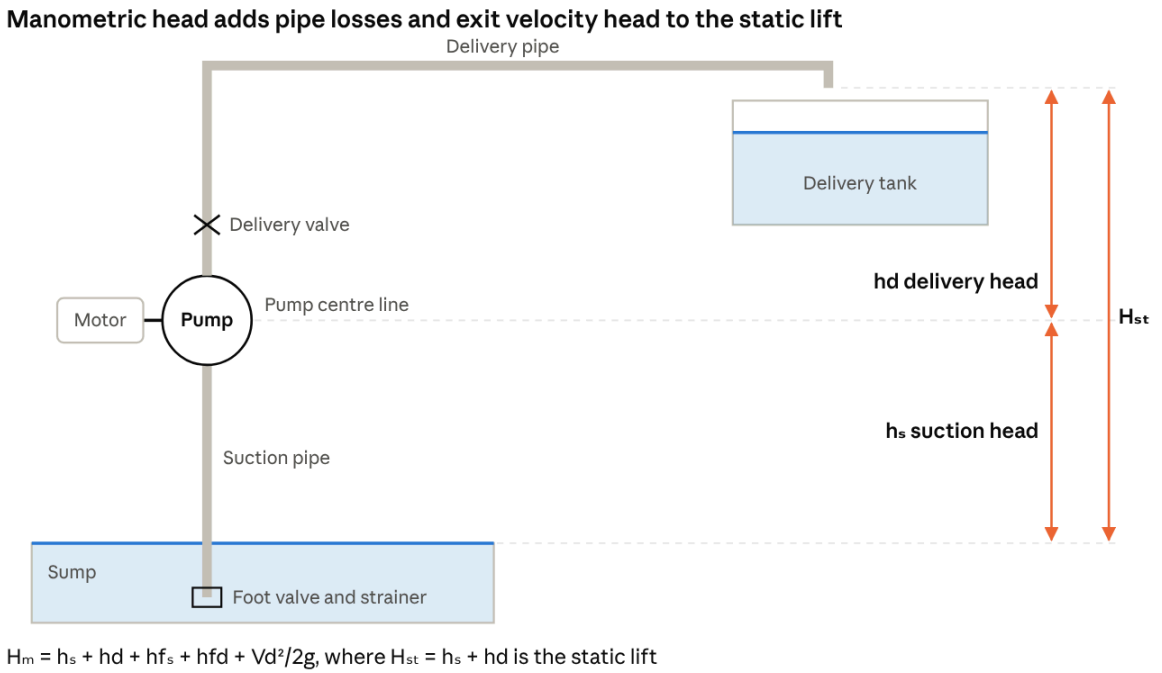


Figure 1. Heads in a centrifugal pump installation

Worked examples

Step-by-step method.

1. List the data in SI units: heads in m, Q in $\mathrm{m^3/s}$.
2. Find the delivery velocity head $V_d^2/(2g)$, if V_d is given.
3. Add all heads: $H_m = h_s + h_d + h_{fs} + h_{fd} + V_d^2/(2g)$.
4. Find the water power $\rho g Q H_m$ in W, and divide by 1000 for kW.
5. Divide by η_o to get the shaft power P.
6. If asked, split $\eta_o = \eta_{\text{mano}}\,\eta_m$ and check that each efficiency is below 1.

Example 2.0 (warm-up) CORE A pump lifts water from a sump 3 m below the pump centre line to a tank 17 m above it. The friction head is 0.4 m in the suction pipe and 1.6 m in the delivery pipe. Water leaves the delivery pipe at 2 m/s. The discharge is $0.04\ \mathrm{m^3/s}$ and $\eta_o = 80\%$. Find H_m and the shaft power.

1. $\dfrac{V_d^2}{2g} = \dfrac{2^2}{2\times 9.81} = 0.204\ \mathrm{m}$
2. $H_m = h_s + h_d + h_{fs} + h_{fd} + \dfrac{V_d^2}{2g} = 3 + 17 + 0.4 + 1.6 + 0.204 = 22.20\ \mathrm{m}$
3. $P = \dfrac{\rho g Q H_m}{1000\,\eta_o} = \dfrac{1000\times 9.81\times 0.04\times 22.20}{1000\times 0.80} = 10.89\ \mathrm{kW}$

Common mistakes and exam tips.

- Taking the suction lift h_s as positive when the pump is below the sump (flooded suction). In that case h_s is negative.
- Leaving out the friction heads h_{fs}, h_{fd} or the velocity head $V_d^2/(2g)$ when the problem gives them.
- Multiplying by η_o instead of dividing. Shaft power is always larger than water power.
- Using Q in L/s or $\mathrm{m^3/h}$ without converting to $\mathrm{m^3/s}$.
- Confusing η_{mano} with η_m. The manometric efficiency compares heads; the mechanical efficiency compares powers.

Check your understanding.

1. CORE A pump has $h_s = 4\ \mathrm{m}$ and $h_d = 26\ \mathrm{m}$. What is the static head? *(It is $30\ \mathrm{m}$, since $H_{\text{st}} = h_s + h_d$.)*
2. CORE If $\eta_{\text{mano}} = 0.80$ and $\eta_m = 0.95$, what is η_o? *(It is $0.80 \times 0.95 = 0.76$.)*
3. HOTS Why must the motor power be more than the water power? *(Hydraulic, mechanical and leakage losses mean $\eta_o < 1$, so $P = \text{water power}/\eta_o$.)*

3. Vector diagrams and work done by the impeller (L3) CORE

The work done by the impeller on each newton of liquid is $V_{w2}u_2/g$, because the liquid enters the impeller radially with no whirl. This is Euler's equation of Unit IV with the flow reversed.

IN SIMPLE WORDS

The impeller gives the water two things: speed along the radius and swirl around the wheel. Imagine a ball sliding outward along a spoke of a spinning wheel. It leaves the rim with a sideways speed, and that sideways speed carries the energy. The work done on each newton of water depends on the sideways (whirl) speed V_{w2} and on the blade speed u_2. Velocity triangles are the drawings that let us find these speeds.

3.1 Velocity triangles CORE

IN SIMPLE WORDS

A velocity triangle shows three speeds of the water on a vane. The blade speed u is how fast the vane moves. The relative speed V_r is how fast the water slides along the vane. The absolute speed V is what an observer beside the pump sees, and it is the sum of the first two. Draw the triangle at the inlet and at the outlet, and read off the whirl part V_w and the flow part V_f from it.

Inlet (eye of impeller). The liquid enters radially, so the absolute velocity has no whirl component.

$$u_1 = \frac{\pi D_1 N}{60}, \qquad V_{w1} = 0, \qquad V_1 = V_{f1}, \qquad \tan\theta = \frac{V_{f1}}{u_1}$$

Outlet (periphery of impeller). The vane leaves at angle ϕ measured back from the direction of rotation.

$$u_2 = \frac{\pi D_2 N}{60}, \qquad V_{w2} = u_2 - \frac{V_{f2}}{\tan\phi}, \qquad \tan\phi = \frac{V_{f2}}{u_2 - V_{w2}}$$

Continuity through the impeller. With impeller widths B_1 and B_2 (vane thickness neglected):

$$Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$$

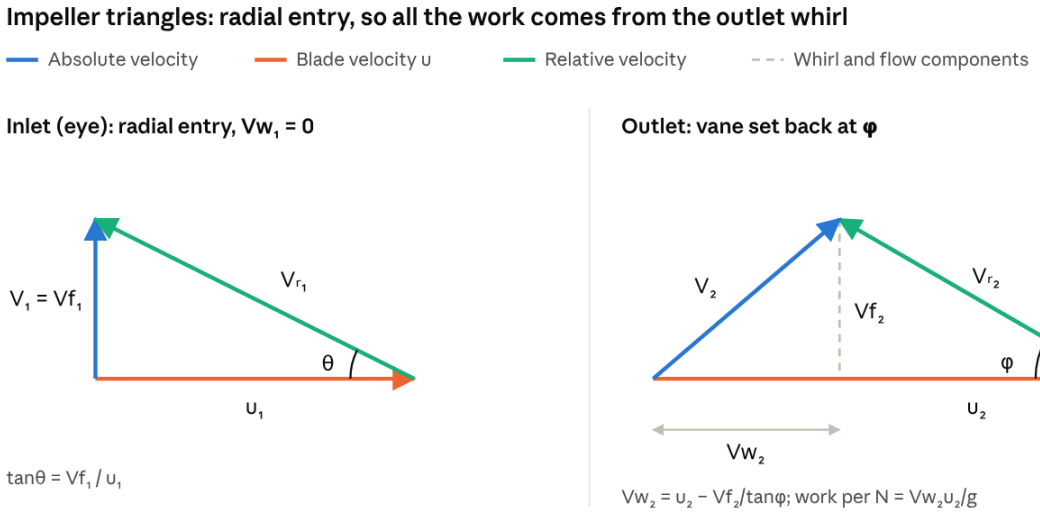


Figure 2. Impeller inlet and outlet velocity triangles

The outlet triangle is drawn for backward-curved vanes ($\phi < 90^\circ$), so $V_{w2} < u_2$ (diagram not to scale).

3.2 Work done by the impeller (derivation) CORE

IN SIMPLE WORDS

A spinning vane pushes the water sideways and so gives it angular momentum, like a skater spinning a ball on a string. The push needed to do this is the torque. Torque times angular speed is the power. Divide that power by the weight of water per second, and you get the work done on each newton of water. For radial entry the answer is $V_{w2}u_2/g$, which is the head the impeller would give if there were no losses.

Derivation from scratch: work done by the impeller (Euler's equation for a pump)

Aim. Prove that the impeller does $V_{w2}u_2/g$ of work on each newton of liquid, when the liquid enters the impeller without whirl.

Assumptions.

1. The flow is steady, the liquid is incompressible, and the discharge through the impeller is Q.
2. The velocities are the same all round the circumference at the inlet radius r_1 and at the outlet radius r_2.
3. Losses are ignored, so we find the ideal (theoretical) head.
4. The impeller turns at constant angular velocity ω, so $u = \omega r$.

Step 1. Angular momentum carried by the liquid. The mass of liquid passing through per second is ρQ. Only the whirl component V_w has a moment about the axis, because the flow component points along the radius and passes through the axis. The moment of momentum entering per second is $\rho Q V_{w1} r_1$. The moment of momentum leaving per second is $\rho Q V_{w2} r_2$.

Step 2. Torque. By Newton's second law for rotation, the torque on the liquid equals the rate of change of angular momentum:

$$T = \rho Q \left(V_{w2} r_2 - V_{w1} r_1 \right)$$

Step 3. Power given to the liquid. Work per second is torque times angular velocity. Then put $u_1 = \omega r_1$ and $u_2 = \omega r_2$:

$$\text{Power} = T\omega = \rho Q \left(V_{w2}\, \omega r_2 - V_{w1}\, \omega r_1 \right) = \rho Q \left(V_{w2} u_2 - V_{w1} u_1 \right)$$

Step 4. Divide by the weight flow. The weight of liquid passing per second is $W = \rho g Q$. The work done per unit weight is

$$\frac{\text{Power}}{\rho g Q} = \frac{\rho Q \left(V_{w2} u_2 - V_{w1} u_1 \right)}{\rho g Q} = \frac{V_{w2} u_2 - V_{w1} u_1}{g}$$

Step 5. Radial entry. The liquid enters the eye with no whirl, so $V_{w1} = 0$:

$$\text{Work done per unit weight} = \frac{V_{w2} u_2}{g}$$

Step 6. Work per second. Multiply by the weight flow $W = \rho g Q$:

$$\text{Work done per second} = \frac{W}{g} V_{w2} u_2 = \frac{\rho g Q}{g} V_{w2} u_2 = \rho Q V_{w2} u_2$$

Result.

$$\text{Work done per unit weight} = \frac{V_{w2} u_2}{g}$$

$$\text{Work done per second} = \frac{W}{g} V_{w2} u_2 = \rho Q V_{w2} u_2$$

What it tells us. The head given by the impeller grows with the blade speed u_2 and with the whirl velocity V_{w2} that the vanes give the liquid. The result does not depend on the shape of the vanes in between, only on the velocities at the inlet and the outlet. It is Euler's equation of Unit IV with the flow reversed.

The term $V_{w2}u_2/g$ is also called the **Euler head** or theoretical head. The manometric head is always smaller because of hydraulic losses: $H_m = \eta_{\text{mano}} \times V_{w2}u_2/g$.

3.3 Effect of outlet vane angle HOTS

IN SIMPLE WORDS

The outlet vane angle ϕ decides how the head changes when you open the delivery valve. With backward-curved vanes ($\phi < 90^\circ$), more flow means a little less head, which is stable and safe. Forward-curved vanes do the opposite: more flow gives more head and more power, so the pump can overload and run unstably. That is why almost all pumps use backward-curved vanes.

From the outlet triangle, $V_{w2} = u_2 - V_{f2}\cot\phi$, and $V_{f2} = Q/(\pi D_2 B_2)$. So the theoretical head depends on discharge as:

$$H_{th} = \frac{u_2^2}{g} - \frac{u_2 \cot\phi}{g\pi D_2 B_2} Q$$

Derivation from scratch: theoretical head against discharge

Aim. Prove that $H_{th} = \dfrac{u_2^2}{g} - \dfrac{u_2 \cot\phi}{g\pi D_2 B_2} Q$.

Assumptions.

1. Radial entry, so $V_{w1} = 0$ and the head is $H_{th} = V_{w2}u_2/g$ (Section 3.2).
2. The vane thickness is neglected, and the flow is uniform around the rim.

Step 1. Outlet triangle. The relative velocity V_{r2} points along the vane at angle ϕ to the reversed blade velocity. Its component along the blade velocity is $V_{r2}\cos\phi$ and its flow component is $V_{r2}\sin\phi = V_{f2}$. Dividing,

$$V_{r2}\cos\phi = V_{f2}\cot\phi$$

Step 2. Whirl velocity. The blade velocity is the sum of the whirl velocity and the part of V_{r2} along u_2:

$$u_2 = V_{w2} + V_{r2}\cos\phi \quad \Rightarrow \quad V_{w2} = u_2 - V_{f2}\cot\phi$$

Step 3. Flow velocity from continuity. The flow area at the rim is $\pi D_2 B_2$:

$$Q = \pi D_2 B_2 V_{f2} \quad \Rightarrow \quad V_{f2} = \frac{Q}{\pi D_2 B_2}$$

Step 4. Substitute into the Euler head.

$$H_{th} = \frac{V_{w2}u_2}{g} = \frac{u_2\,(\,u_2 - V_{f2}\cot\phi\,)}{g} = \frac{u_2^2}{g} - \frac{u_2\cot\phi}{g}\cdot\frac{Q}{\pi D_2 B_2}$$

Result.

$$H_{th} = \frac{u_2^2}{g} - \frac{u_2 \cot \phi}{g \pi D_2 B_2} Q$$

What it tells us. The graph of H_{th} against Q is a straight line with slope $-u_2 \cot \phi / (g \pi D_2 B_2)$. For $\phi < 90^\circ$, $\cot \phi$ is positive and the line falls. For $\phi = 90^\circ$, $\cot \phi = 0$ and the line is flat at u_2^2/g. For $\phi > 90^\circ$, $\cot \phi$ is negative and the line rises.

Vane shape	Outlet angle ϕ	Theoretical head as Q rises	Use
Backward-curved	$\phi < 90^\circ$	Falls	Almost all pumps: stable operation, highest efficiency, power does not overload
Radial	$\phi = 90^\circ$	Stays constant at u_2^2/g	Rare; simple to make
Forward-curved	$\phi > 90^\circ$	Rises	Avoided in pumps: unstable, power rises steeply with flow; used in some fans

Worked examples

Step-by-step method.

1. Convert diameters to m and speed to rpm. Find $u = \pi D N/60$ at inlet and outlet.
2. Find the flow velocity from continuity, $V_f = Q/ (\pi D B)$, if it is not given.
3. Draw the outlet triangle. Get $V_{w2} = u_2 - V_{f2}/ \tan \phi$.
4. Find the work per unit weight, $V_{w2} u_2 / g$ (radial entry, $V_{w1} = 0$).
5. Find $H_m = \eta_{\text{mano}} V_{w2} u_2 / g$, or $\eta_{\text{mano}} = g H_m / (V_{w2} u_2)$.
6. Find the power with $P = \rho g Q H_m / \eta_o$, and check that $\eta_m = \eta_o / \eta_{\text{mano}}$ is below 1.

Example 3.0 (warm-up) CORE An impeller of outer diameter 200 mm runs at 1500 rpm. The flow velocity at the outlet is 2 m/s and the outlet vane angle is 45°. Find the work done per unit weight.

1. $u_2 = \dfrac{\pi D_2 N}{60} = \dfrac{\pi \times 0.2 \times 1500}{60} = 15.71 \text{ m/s}$
2. $V_{w2} = u_2 - \dfrac{V_{f2}}{\tan \phi} = 15.71 - \dfrac{2}{\tan 45^\circ} = 13.71 \text{ m/s}$
3. Work per unit weight $= \dfrac{V_{w2} u_2}{g} = \dfrac{13.71 \times 15.71}{9.81} = 21.95 \text{ N} \cdot \text{m/N}$

Example 3.1 CORE A centrifugal pump impeller has outer and inner diameters 300 mm and 150 mm and runs at 1450 rpm. The vanes are set back at 30° at the outlet and the velocity of flow is constant at 2.5 m/s. Find the vane angle at inlet, the work done per unit weight and the manometric head if $\eta_{\text{mano}} = 80\%$.

1. $u_1 = \dfrac{\pi \times 0.15 \times 1450}{60} = 11.39\ \text{m/s};\ u_2 = \dfrac{\pi \times 0.30 \times 1450}{60} = 22.78\ \text{m/s}$
2. Inlet: $\tan\theta = \dfrac{V_{f1}}{u_1} = \dfrac{2.5}{11.39}$, so $\theta = 12.4^\circ$
3. Outlet: $V_{w2} = u_2 - \dfrac{V_{f2}}{\tan\phi} = 22.78 - \dfrac{2.5}{\tan 30^\circ} = 22.78 - 4.33 = 18.45\ \text{m/s}$
4. Work per unit weight $= \dfrac{V_{w2}u_2}{g} = \dfrac{18.45 \times 22.78}{9.81} = 42.8\ \text{N·m/N}$
5. $H_m = 0.8 \times 42.8 = 34.3\ \text{m}$

Example 3.2 HOTS A pump with impeller outer diameter 400 mm and outlet width 50 mm runs at 1000 rpm and delivers $0.08\ \text{m}^3/\text{s}$ against a manometric head of 30 m. The outlet vane angle is 35°. Find the manometric efficiency and the shaft power if the overall efficiency is 70%.

1. $V_{f2} = \dfrac{Q}{\pi D_2 B_2} = \dfrac{0.08}{\pi \times 0.4 \times 0.05} = 1.273\ \text{m/s}$
2. $u_2 = \dfrac{\pi \times 0.4 \times 1000}{60} = 20.94\ \text{m/s}$
3. $V_{w2} = 20.94 - \dfrac{1.273}{\tan 35^\circ} = 20.94 - 1.82 = 19.13\ \text{m/s}$
4. $\eta_{\text{mano}} = \dfrac{gH_m}{V_{w2}u_2} = \dfrac{9.81 \times 30}{19.13 \times 20.94} = 73.5\%$
5. $P = \dfrac{\rho g Q H_m}{\eta_o} = \dfrac{1000 \times 9.81 \times 0.08 \times 30}{0.70} = 33.6\ \text{kW}$
6. Check: $\eta_m = \dfrac{\eta_o}{\eta_{\text{mano}}} = \dfrac{0.70}{0.735} = 0.95$, a realistic mechanical efficiency

Common mistakes and exam tips.

- Measuring the vane angle from the wrong direction. Here ϕ is measured from the reversed blade velocity, so $\tan\phi = V_{f2}/(u_2 - V_{w2})$, not V_{f2}/V_{w2}.
- Forgetting that $V_{w1} = 0$ holds only for radial entry. If the problem gives an inlet whirl, use $(V_{w2}u_2 - V_{w1}u_1)/g$.
- Using the diameter in mm or the speed in rad/s inside $u = \pi DN/60$. Use metres and rpm.
- Using η_o to find H_m from the Euler head. Use η_{mano}: $H_m = \eta_{\text{mano}} V_{w2}u_2/g$.
- Taking the flow velocity as constant when it is not. Use $V_f = Q/(\pi DB)$ at each radius.

Check your understanding.

1. CORE What is V_{w1} for radial entry? *(It is zero.)*

2. CORE For $u_2 = 20\,\mathrm{m/s}$ and $V_{w2} = 15\,\mathrm{m/s}$, what is the work per unit weight? *(It is $15 \times 20/9.81 = 30.6\,\mathrm{m}$.)*
3. HOTS For backward-curved vanes, what happens to the theoretical head as the discharge rises at constant speed? *(It falls, because V_{f2} rises and so $V_{w2} = u_2 - V_{f2}\cot\phi$ falls.)*

4. Starting speed, multistage pumps, specific speed and model laws (L4) CORE

A centrifugal pump delivers only when its centrifugal head exceeds the manometric head, so it has a minimum starting speed. Specific speed then classifies the impeller shape, and the model laws scale performance from one speed or size to another.

> **IN SIMPLE WORDS**
> A pump cannot lift water unless it spins fast enough. If the speed is too low, the spinning water cannot build the pressure to beat the head, and nothing comes out. Once the pump works, you may need more head than one wheel gives, so you put wheels in series. Specific speed is a single number that tells the shape of an impeller, like a clothes size. The affinity laws then predict what happens when the speed changes.

4.1 Minimum starting speed CORE

> **IN SIMPLE WORDS**
> Before the pump delivers, the impeller only spins the water inside it. The water at the rim is then at a higher pressure than the water at the eye, like water pressed outward in a spinning bucket. If this pressure rise is smaller than the head to be overcome, no water leaves the pump. The minimum starting speed is the speed at which the rise just equals the head H_m.

At starting, before flow is established, the only head the impeller produces is the centrifugal (forced-vortex) head. Flow begins when this equals the manometric head:

$$\frac{u_2^2 - u_1^2}{2g} = H_m \quad \Rightarrow \quad N_{\min} = \frac{60}{\pi}\sqrt{\frac{2gH_m}{D_2^2 - D_1^2}}$$

Derivation from scratch: minimum starting speed

Aim. Prove that $N_{\min} = \dfrac{60}{\pi}\sqrt{\dfrac{2gH_m}{D_2^2 - D_1^2}}$.

Assumptions.

1. At starting there is no flow, so the liquid in the impeller turns as a solid body (forced vortex).
2. Flow begins when the pressure-head rise between the eye and the rim just equals the manometric head H_m.
3. D_1 and D_2 are the inner (eye) and outer (rim) diameters, and N is in rpm.

Step 1. Pressure-head rise. From the forced-vortex result of Section 1:

$$\text{Rise in head} = \frac{u_2^2 - u_1^2}{2g}$$

Step 2. Starting condition. Flow begins when the rise equals H_m:

$$\frac{u_2^2 - u_1^2}{2g} = H_m$$

Step 3. Put in the blade velocities. Use $u_1 = \pi D_1 N/60$ and $u_2 = \pi D_2 N/60$:

$$\left(\frac{\pi N}{60}\right)^2 \frac{D_2^2 - D_1^2}{2g} = H_m$$

Step 4. Solve for N. Multiply by $2g$ and divide by $D_2^2 - D_1^2$, then take the square root:

$$\left(\frac{\pi N}{60}\right)^2 = \frac{2gH_m}{D_2^2 - D_1^2} \quad \Rightarrow \quad \frac{\pi N}{60} = \sqrt{\frac{2gH_m}{D_2^2 - D_1^2}}$$

Step 5. Isolate the speed.

$$N_{\min} = \frac{60}{\pi}\sqrt{\frac{2gH_m}{D_2^2 - D_1^2}}$$

Second form. Put $H_m = \eta_{\text{mano}} V_{w2} u_2 / g$ into Step 2, so that $u_2^2 - u_1^2 = 2\,\eta_{\text{mano}} V_{w2} u_2$:

$$\left(\frac{\pi N}{60}\right)^2 \left(D_2^2 - D_1^2\right) = 2\,\eta_{\text{mano}} V_{w2} \left(\frac{\pi N}{60}\right) D_2$$

$$\Rightarrow \quad \frac{\pi N}{60}\left(D_2^2 - D_1^2\right) = 2\,\eta_{\text{mano}} V_{w2} D_2$$

$$N_{\min} = \frac{120\,\eta_{\text{mano}}\, V_{w2}\, D_2}{\pi\left(D_2^2 - D_1^2\right)}$$

Result.

$$N_{\min} = \frac{60}{\pi}\sqrt{\frac{2gH_m}{D_2^2 - D_1^2}}$$

What it tells us. A higher manometric head needs a higher starting speed, because $N_{\min} \propto \sqrt{H_m}$. A larger rim diameter or a smaller eye diameter lowers it, because the pressure rise is larger. A pump with a high static lift must be run well above $N_{\min}$.

Step-by-step method.

1. Convert diameters to m and heads to m. Note whether the question asks for the speed in rpm.
2. For the starting speed, use $N_{\min} = \dfrac{60}{\pi}\sqrt{\dfrac{2gH_m}{D_2^2 - D_1^2}}$, with the difference of the squares of the diameters.
3. For stages, divide the head required by the head of one impeller and round up to a whole number.
4. For parallel pumps, divide the discharge required by the discharge of one pump.
5. For specific speed use $N_s = N\sqrt{Q}\,/H_m^{3/4}$ with N in rpm, Q in $\mathrm{m^3/s}$ and H_m in m. For a multistage pump use the head of one stage.
6. Compare N_s with the table to name the impeller type.

Example 4.0 (warm-up) CORE An impeller has outer and inner diameters 400 mm and 200 mm and works against a manometric head of 15 m. Find the minimum starting speed.

1. $D_2^2 - D_1^2 = 0.4^2 - 0.2^2 = 0.16 - 0.04 = 0.12\ \mathrm{m^2}$
2. $\dfrac{\pi N}{60} = \sqrt{\dfrac{2 \times 9.81 \times 15}{0.12}} = \sqrt{2452.5} = 49.52\ \mathrm{rad/s}$
3. $N_{\min} = \dfrac{60}{\pi} \times 49.52 = 946\ \mathrm{rpm}$

Example 4.1 CORE A pump has impeller diameters 500 mm (outer) and 250 mm (inner) and works against a manometric head of 20 m. Find the minimum starting speed.

1. $\left(\dfrac{\pi N}{60}\right)^2 \dfrac{D_2^2 - D_1^2}{2g} = H_m$, so $\left(\dfrac{\pi N}{60}\right)^2 \times (\,0.25 - 0.0625\,) = 2 \times 9.81 \times 20 = 392.4$
2. $\left(\dfrac{\pi N}{60}\right)^2 = 2092.8$, so $\dfrac{\pi N}{60} = 45.75\ \mathrm{rad/s}$
3. $N_{\min} = 874\ \mathrm{rpm}$

4.2 Multistage pumps CORE

IN SIMPLE WORDS

One impeller can give only a limited head. For a very high head, put several impellers on one shaft, so the water passes from one to the next, like climbing a staircase one step at a time. For a very large flow, run several pumps side by side, like opening several taps into one tank. Series adds head. Parallel adds discharge.

Arrangement	What adds	Total	Use
n impellers in series on one shaft	Head	Total head $= n \times H_m$, discharge $= Q$	High heads, e.g. boiler feed, deep wells
n pumps in parallel	Discharge	Total discharge $= n \times Q$, head $= H_m$	Large flows at moderate head, e.g. city water supply

Derivation from scratch: head and discharge of multistage and parallel pumps

Aim. Show that n impellers in series give nH_m at discharge Q, and that n pumps in parallel give nQ at head H_m.

Assumptions.

1. All impellers or pumps are identical and each develops the head H_m at the discharge Q passing through it.
2. Losses in the connecting passages are included in H_m.

Step 1. Series arrangement. The delivery of stage 1 is the suction of stage 2, and so on. The same liquid, and so the same discharge Q, passes through every impeller (continuity). If the liquid enters stage 1 with total head H_0, each stage raises the total head by H_m:

$$H_0 \;\rightarrow\; H_0 + H_m \;\rightarrow\; H_0 + 2H_m \;\rightarrow\; \cdots \;\rightarrow\; H_0 + nH_m$$

The head added by the whole pump is therefore

$$\text{Total head} = nH_m, \qquad \text{discharge} = Q$$

Step 2. Parallel arrangement. All pumps draw from the same sump and deliver into the same delivery main. Every pump therefore works between the same suction and delivery pressures, so every pump works at the same head H_m. At the junction of the delivery pipes the discharges add (continuity):

$$\text{Total discharge} = Q + Q + \cdots + Q = nQ, \qquad \text{head} = H_m$$

Step 3. Number of stages for a required head. If H is the total head required, then $nH_m = H$, so $n = H/H_m$ (rounded up to a whole number).

Result.

$$\text{Series: } nH_m \text{ at } Q \qquad \text{Parallel: } nQ \text{ at } H_m$$

What it tells us. Series suits a high head with a moderate flow. Parallel suits a large flow with a moderate head. The total water power $\rho g Q H_m n$ is the same in both cases, but the pipes and pump sizes differ.

Example 4.2 CORE A head of 120 m is required and one impeller can develop 40 m at the required discharge. Number of stages $= 120/40 =$ 3 impellers in series.

4.3 Specific speed of a pump CORE

IN SIMPLE WORDS

Pumps come in many sizes, so we need a single number that tells the type of impeller. Specific speed is that number. Imagine shrinking or enlarging a pump until it delivers $1\ \mathrm{m}^3/\mathrm{s}$ against a head of $1\ \mathrm{m}$. The speed it would then need is N_s. A small N_s means a high head and a small flow, and a large N_s means a large flow and a low head.

The specific speed N_s of a pump is the speed of a geometrically similar pump that delivers $1\ \mathrm{m}^3/\mathrm{s}$ against a head of $1\ \mathrm{m}$.

Derivation from scratch: specific speed of a pump

Aim. Prove that $N_s = N\sqrt{Q}\,/H_m^{3/4}$.

Assumptions.

1. The pumps are geometrically similar, so their velocity triangles are similar. Then every velocity is proportional to the blade velocity u.
2. The efficiencies are the same, and g is constant.
3. N_s is the speed of a similar pump that delivers $1\ \mathrm{m}^3/\mathrm{s}$ against a head of $1\ \mathrm{m}$.

Step 1. Discharge. The discharge is the flow area times the flow velocity. The flow area πDB is proportional to D^2 for similar pumps, because $B \propto D$. The flow velocity V_f is proportional to u, and $u = \pi DN/60 \propto DN$:

$$Q \propto D^2 V_f \propto D^2 \cdot DN = D^3 N$$

Step 2. Head. The head is proportional to u^2/g, so $u \propto \sqrt{H_m}$. Since $u \propto DN$:

$$DN \propto \sqrt{H_m} \quad \Rightarrow \quad D \propto \frac{\sqrt{H_m}}{N}$$

Step 3. Eliminate D. Put $D^3 \propto H_m^{3/2}/N^3$ from Step 2 into Step 1:

$$Q \propto \frac{H_m^{3/2}}{N^3} N = \frac{H_m^{3/2}}{N^2}$$

Step 4. Form a constant. Rearranging Step 3, the group below is the same for all similar pumps:

$$\frac{N^2 Q}{H_m^{3/2}} = \text{constant}$$

Step 5. Fix the constant. By definition, when $Q = 1\ \mathrm{m}^3/\mathrm{s}$ and $H_m = 1\ \mathrm{m}$, the speed is $N = N_s$. The constant is therefore N_s^2:

$$N_s^2 = \frac{N^2 Q}{H_m^{3/2}}$$

Step 6. Take the square root. Note $\left(H_m^{3/2}\right)^{1/2} = H_m^{3/4}$:

$$N_s = \frac{N\sqrt{Q}}{H_m^{3/4}}$$

Result.

$$N_s = \frac{N\sqrt{Q}}{H_m^{3/4}}$$

What it tells us. A high N_s means a large discharge at a low head, which gives a wide mixed-flow or axial impeller. A low N_s means a small discharge at a high head, which gives a narrow radial impeller of large diameter. The number lets us compare pumps of different size on equal terms.

The units to use in this result are:

$$N_s = \frac{N\sqrt{Q}}{H_m^{3/4}} \qquad \left(N \text{ in rpm},\ Q \text{ in } \mathrm{m^3/s},\ H_m \text{ in m}\right)$$

Impeller type	N_s (rpm, $\mathrm{m^3/s}$, m)	Shape
Radial flow, slow	10–30	Narrow, large diameter
Radial flow, medium	30–50	
Radial flow, fast	50–80	Wider outlet
Mixed flow	80–160	Flow leaves at an angle
Axial flow (propeller pump)	160–500	Propeller in a pipe

Note the difference from a turbine: a turbine's specific speed uses power P, a pump's uses discharge Q, because Q is the pump's useful output.

Example 4.3 CORE A pump runs at 1450 rpm and delivers $0.05\ \mathrm{m^3/s}$ against 25 m. $N_s = \dfrac{1450 \times \sqrt{0.05}}{25^{0.75}} = \dfrac{1450 \times 0.2236}{11.18} = 29.0$, a slow-to-medium radial-flow impeller.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

4.4 Model laws (affinity laws) ADVANCED

For geometrically similar pumps (or the same pump at different speeds), the dimensionless groups below stay constant:

$$\frac{Q}{ND^3} = \text{const}, \qquad \frac{gH_m}{N^2D^2} = \text{const}, \qquad \frac{P}{N^3D^5} = \text{const}$$

For the same pump (D fixed) these reduce to $Q \propto N$, $H_m \propto N^2$ and $P \propto N^3$.

Example 4.4 ADVANCED The pump of Example 3.2 ($0.08\,\mathrm{m^3/s}$, 30 m, 33.6 kW at 1000 rpm) is run at 1200 rpm. $Q = 0.08 \times 1.2 = 0.096\,\mathrm{m^3/s}$, $H_m = 30 \times 1.2^2 = 43.2\,\mathrm{m}$, $P = 33.6 \times 1.2^3 = 58.1\,\mathrm{kW}$.

4.5 Characteristic curves ADVANCED

Curve	How it is obtained	What it shows
Main characteristics	Speed held constant at several values; Q varied with the delivery valve	H_m, P and η_o against Q at each speed
Operating characteristics	Speed held at the design value	H_m, P and η_o against Q; the design point is at peak η_o
Constant-efficiency (iso-efficiency) curves	Cross-plotted from the main characteristics	H_m against Q with lines of equal efficiency; shows the best operating zone

With backward-curved vanes, H_m falls as Q rises, power rises to a peak and then levels off, and efficiency has a single maximum at the design discharge.

Common mistakes and exam tips.

- Using diameters in mm in $N_{\min}$. Convert to m, and take the difference of the squares, $D_2^2 - D_1^2$, not $(D_2 - D_1)^2$.
- Forgetting to round the number of stages up to a whole number.
- Mixing the two arrangements. Series adds head; parallel adds discharge.
- Using the turbine formula (with power) for a pump. The pump specific speed uses $\sqrt{Q}$ and $H_m^{3/4}$.
- Using the total head of a multistage pump in N_s. Use the head of one stage.

Check your understanding.

1. CORE Two identical impellers on one shaft run in series. What happens to the head and the discharge? *(The head doubles; the discharge stays the same.)*
2. CORE The head to be overcome becomes four times as large. How does $N_{\min}$ change? *(It doubles, since $N_{\min} \propto \sqrt{H_m}$.)*
3. HOTS Pump A has $N_s = 25$ and pump B has $N_s = 120$. Which gives the larger discharge for the same head and speed? *(Pump B. Pump A is a slow radial impeller for high head; pump B is a mixed-flow impeller for larger discharge at lower head.)*

5. Cavitation and separation (L5) CORE

Cavitation starts wherever the local pressure of a flowing liquid falls to its vapour pressure. In a centrifugal pump that point is the eye of the impeller, the lowest-pressure point in the installation; in a reciprocating pump it is the start of the suction stroke (see 7.4).

IN SIMPLE WORDS

Water boils at room temperature if the pressure is low enough. Inside a pump the pressure is lowest at the eye of the impeller. If it drops to the vapour pressure, bubbles of vapour form. When the bubbles reach higher pressure they collapse violently and hammer the metal. Engineers check this with NPSH, which is the pressure margin above boiling at the pump inlet. The factor σ is the same margin written as a fraction of the pump head.

5.1 Mechanism CORE

IN SIMPLE WORDS

Cavitation is boiling caused by low pressure instead of heat. Think of the pressure falling as the water speeds up into the eye. When it reaches the vapour pressure, vapour bubbles appear. The bubbles are swept to a place of higher pressure and collapse suddenly. Each collapse is a tiny hammer blow on the vane, repeated thousands of times a second.

1. As the liquid accelerates into the eye, its pressure falls. If it reaches the vapour pressure p_v, the liquid boils locally and vapour bubbles (cavities) form; dissolved gases also come out of solution.
2. The bubbles are carried into the vane passages, where the pressure is higher.
3. There they collapse suddenly. The surrounding liquid rushes in, and the collapse produces local pressure spikes reported to reach several thousand atmospheres, repeated thousands of times a second on the same spot.

5.2 Effects CORE

- **Pitting and erosion** of the vane surfaces near the inlet, which can destroy an impeller over time
- **Noise and vibration**, often described as a sound like gravel passing through the pump
- **Fall in head, discharge and efficiency**, because the vapour pockets block part of the flow passage

5.3 Net positive suction head (NPSH) CORE

IN SIMPLE WORDS

NPSH is the safety margin against boiling at the pump inlet. It is the total head of the liquid at the inlet minus the head at which the liquid would boil. Think of the atmosphere pushing water up the suction pipe. The atmosphere gives about 10.3 m of head, but lifting the water, friction in the pipe and the boiling head all use some of it. What remains is NPSH. The pump needs a certain margin, and the installation must supply at least that much.

NPSH is the head available at the pump inlet above the vapour pressure head. For a pump drawing from an open sump:

$$\text{NPSH} = \frac{p_1}{\rho g} + \frac{V_1^2}{2g} - \frac{p_v}{\rho g} = H_{\text{atm}} - H_v - h_s - h_{fs}$$

where H_{atm} = atmospheric pressure head (about 10.3 m of water at sea level), H_v = vapour pressure head (about 0.24 m for water at 20 °C), h_s = suction lift and h_{fs} = friction loss in the suction pipe.

Derivation from scratch: NPSH of a pump drawing from an open sump

Aim. Prove that $\text{NPSH} = H_{\text{atm}} - H_v - h_s - h_{fs}$.

Assumptions.

1. Steady flow of an incompressible liquid from a large open sump.
2. The sump surface is at atmospheric pressure p_{atm} and its velocity is negligible.
3. Section 1 is the pump inlet, a height h_s above the sump surface. h_{fs} is the friction (and fitting) loss in the suction pipe.
4. $H_{\text{atm}} = p_{\text{atm}}/\rho g$ and $H_v = p_v/\rho g$.

Step 1. Energy equation from the sump surface to the pump inlet. Take the sump surface as the datum:

$$\frac{p_{\text{atm}}}{\rho g} + 0 + 0 = \frac{p_1}{\rho g} + \frac{V_1^2}{2g} + h_s + h_{fs}$$

Step 2. Total head at the inlet. Move h_s and h_{fs} to the left:

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} = H_{\text{atm}} - h_s - h_{fs}$$

Step 3. Subtract the vapour head. NPSH is the total head at the inlet above the vapour pressure head:

$$\text{NPSH} = \frac{p_1}{\rho g} + \frac{V_1^2}{2g} - \frac{p_v}{\rho g} = H_{\text{atm}} - h_s - h_{fs} - H_v$$

Result.

$$\text{NPSH} = \frac{p_1}{\rho g} + \frac{V_1^2}{2g} - \frac{p_v}{\rho g} = H_{\text{atm}} - H_v - h_s - h_{fs}$$

What it tells us. Every metre of suction lift or friction loss lowers NPSH by one metre. A higher vapour pressure (hot liquid) also lowers it. A lower atmospheric pressure (high altitude) lowers it too. For a flooded suction, h_s is negative and NPSH rises.

- **NPSH available (NPSHA)** — fixed by the installation (the expression above)
- **NPSH required (NPSHR)** — fixed by the pump design, given by the manufacturer
- Condition to avoid cavitation: $\text{NPSHA} > \text{NPSHR}$, usually with a margin of about 0.5–1 m

5.4 Thoma’s cavitation factor CORE

IN SIMPLE WORDS

NPSH is in metres, so a large pump and a small pump are hard to compare. Thoma’s factor σ divides NPSH by the pump head H_m, which gives a plain number. Every pump type has a critical value σ_c found by test. If the installation has a σ above σ_c, the pump is safe. If not, the suction lift must be reduced.

$$\sigma = \frac{\text{NPSH}}{H_m} = \frac{H_{\text{atm}} - H_v - h_s - h_{fs}}{H_m}$$

Cavitation does not occur as long as σ is above a critical value σ_c, found by test for each pump type. The maximum permissible suction lift follows directly:

$$h_{s,\,\max} = H_{\text{atm}} - H_v - h_{fs} - \sigma_c H_m$$

Derivation from scratch: Thoma’s factor and the maximum suction lift

Aim. Prove that $h_{s,\,\max} = H_{\text{atm}} - H_v - h_{fs} - \sigma_c H_m$.

Assumptions.

1. NPSH is given by the open-sump expression derived in 5.3.
2. Cavitation is just avoided when σ equals the critical value σ_c.

Step 1. Definition. Thoma’s factor is the NPSH divided by the manometric head:

$$\sigma = \frac{\text{NPSH}}{H_m} = \frac{H_{\text{atm}} - H_v - h_s - h_{fs}}{H_m}$$

Step 2. Limiting case. At the limit, $\sigma = \sigma_c$. Multiply by H_m:

$$\sigma_c H_m = H_{\text{atm}} - H_v - h_s - h_{fs}$$

Step 3. Solve for the suction lift. This lift is the largest permitted, so call it $h_{s,\,\max}$:

$$h_{s,\,\max} = H_{\text{atm}} - H_v - h_{fs} - \sigma_c H_m$$

Result.

$$h_{s,\,\max} = H_{\text{atm}} - H_v - h_{fs} - \sigma_c H_m$$

What it tells us. The term $\sigma_c H_m$ is the NPSH the pump requires (the NPSHR). A pump with a larger head, or a larger σ_c, needs more NPSH, so it must be set lower. Any lift below $h_{s,\,\max}$ is safe.

Step-by-step method.

1. Write every head in metres of water: $H_{\text{atm}} \approx 10.3\,\text{m}$ and H_v from the water temperature.
2. Find $\text{NPSH} = H_{\text{atm}} - H_v - h_s - h_{fs}$. Use a negative h_s for a flooded suction.
3. Find Thoma's factor $\sigma = \text{NPSH}/H_m$ and compare it with σ_c.
4. The pump is safe if $\sigma > \sigma_c$, which is the same as $\text{NPSHA} > \text{NPSHR}$.
5. For the limiting suction lift, use $h_{s,\,\max} = H_{\text{atm}} - H_v - h_{fs} - \sigma_c H_m$.
6. If cavitation is predicted, reduce h_s or h_{fs}, or lower the pump.

Example 5.0 (warm-up) CORE A pump draws water from an open sump with a suction lift of 3 m. The suction friction loss is 0.5 m, $H_{\text{atm}} = 10.3\,\text{m}$ and $H_v = 0.24\,\text{m}$. The pump works against $H_m = 30\,\text{m}$. Find the NPSH and Thoma's factor.

1. $\text{NPSH} = H_{\text{atm}} - H_v - h_s - h_{fs} = 10.3 - 0.24 - 3 - 0.5 = 6.56\,\text{m}$
2. $\sigma = \dfrac{\text{NPSH}}{H_m} = \dfrac{6.56}{30} = 0.219$

Example 5.1 HOTS A pump works against $H_m = 40\,\text{m}$ with a suction lift of 4 m and suction friction loss of 0.8 m. Take $H_{\text{atm}} = 10.3\,\text{m}$, $H_v = 0.25\,\text{m}$ and $\sigma_c = 0.10$. Check for cavitation and find the maximum suction lift.

1. $\text{NPSH} = 10.3 - 0.25 - 4 - 0.8 = 5.25\,\text{m}$
2. $\sigma = \dfrac{5.25}{40} = 0.131 > \sigma_c = 0.10$, so **no cavitation**
3. $h_{s,\,\max} = 10.3 - 0.25 - 0.8 - 0.10 \times 40 = 5.25\,\text{m}$, so the pump could be raised by up to 1.25 m more

5.5 Prevention CORE

- Keep the suction lift small, or place the pump below the sump level (flooded suction)
- Use a short, large-diameter suction pipe with few bends, to keep h_{fs} low
- Avoid running far above the design discharge or speed
- Use a double-suction impeller or an inducer to lower the eye velocity
- Make the impeller of cavitation-resistant material (stainless steel, bronze) and keep its surfaces smooth

5.6 Separation CORE

IN SIMPLE WORDS

Separation is cavitation's cousin. Here a long column of liquid, such as the water in a reciprocating pump's suction pipe, is pulled so hard that it tears away from the piston. A vapour-filled gap appears. This happens at about 2.5 m of water absolute, a little above the vapour pressure, because dissolved air comes out of the liquid first.

Separation is the related phenomenon in which the liquid column breaks away from the surface it should follow, leaving a vapour-filled gap. In a pump it happens when the absolute pressure falls to the **separation pressure head**, taken as about 2.5 m of water absolute (somewhat above the vapour pressure because dissolved air comes out first). It matters most in reciprocating pumps, where the acceleration head lowers the pressure at the start of the suction stroke; section 7.4 derives the resulting speed limit.

Common mistakes and exam tips.

- Reversing the cavitation condition. The pump is safe when the NPSH available is larger than the NPSH required, or when $\sigma > \sigma_c$.
- Forgetting the vapour head H_v or the suction friction h_{fs} in the NPSH expression.
- Treating a flooded suction like a lift. When the pump is below the sump, h_s is negative.
- Using the wrong atmospheric head. Use $H_{\text{atm}} \approx 10.3\,\text{m}$ of water, and a lower value at altitude.
- Forgetting that hot water has a higher vapour pressure, which lowers NPSH.

Check your understanding.

1. CORE NPSH available is 6 m and NPSH required is 4 m. Does the pump cavitate? *(No, because the available value is larger than the required value.)*
2. CORE A pump has $H_m = 40\,\text{m}$ and an NPSH of 6 m. What is σ? *(It is $6/40 = 0.15$.)*
3. HOTS The pump is raised 1 m higher above the sump. What happens to NPSHA? *(It falls by 1 m, since h_s rises by 1 m, and a little more if friction rises.)*

6. Reciprocating pump theory (L6) CORE

A reciprocating pump is a positive-displacement pump: each stroke of the piston traps a fixed volume of liquid and pushes it out, whatever the delivery head. It suits small discharges at high heads.

IN SIMPLE WORDS

A reciprocating pump works like a bicycle pump or a syringe. The piston pulls liquid in, then pushes it out. Each stroke moves one fixed volume, so the flow does not depend on how high you lift the water. Engineers use it when a small flow must go to a very high pressure, such as in oil wells and hydraulic presses. The key symbols are the piston area A, the stroke L and the speed N in rpm.

6.1 Main parts and working CORE

IN SIMPLE WORDS

Think of a syringe with two one-way doors. One door lets liquid in from the tank and the other lets liquid out to the pipe. The doors are the suction valve and the delivery valve. When the piston moves back, the inlet door opens. When it moves forward, the outlet door opens. A crank turned by a motor makes the piston move to and fro.

1. **Cylinder with piston (or plunger)**, moved back and forth by a crank and connecting rod driven by a motor
2. **Suction pipe** with a non-return **suction valve** at the cylinder end
3. **Delivery pipe** with a non-return **delivery valve**

Working of a single-acting pump:

1. **Suction stroke** (crank angle 0 to 180°): the piston moves away from the valves, the cylinder pressure drops below atmospheric, the suction valve opens and liquid enters the cylinder. The delivery valve stays closed.
2. **Delivery stroke** (crank angle 180° to 360°): the piston moves back, the cylinder pressure rises, the suction valve closes and the delivery valve opens, forcing liquid up the delivery pipe.

In a **double-acting** pump, liquid acts on both sides of the piston. Each side has its own suction and delivery valves, so there is one suction and one delivery stroke on each side per revolution, and delivery is roughly twice as large and more uniform.

6.2 Discharge CORE

IN SIMPLE WORDS

The discharge is the volume of liquid delivered each second. In one delivery stroke the piston sweeps out a cylinder of volume (area times stroke). Multiply by the number of delivery strokes per second and you have the flow. A single-acting pump delivers once per revolution. A double-acting pump delivers twice, so it gives about twice the flow.

Derivation from scratch: theoretical discharge of a reciprocating pump

Aim. Find the theoretical volume of liquid delivered per second, Q_{th}, for a single-acting and a double-acting pump.

Assumptions.

1. There is no leakage past the piston or valves, and the valves open and close at once.
2. The cylinder fills completely on every suction stroke (no air, no separation).
3. The piston has area $A = \pi D^2/4$ and moves through a stroke $L = 2r$, where r is the crank radius.
4. The crank turns at a steady N rpm. The area of the piston rod is neglected.

Step 1. Volume swept in one stroke. The piston moves a distance L across an area A, so it sweeps a cylinder of volume:

$$V_{\text{stroke}} = A \times L = AL$$

Step 2. Volume delivered per revolution, single-acting. Liquid is pushed out only during the delivery stroke, which happens once in every revolution. So the volume delivered per revolution is AL.

Step 3. Volume delivered per minute. The crank makes N revolutions in one minute.

$$\text{Volume per minute} = AL \times N = ALN$$

Step 4. Convert to volume per second. One minute is 60 seconds, so divide by 60.

$$Q_{\text{th}} = \frac{ALN}{60}$$

Step 5. Double-acting pump. Both faces of the piston push out liquid, so there are two delivery strokes per revolution. The volume per revolution is $2AL$, and the same steps 3 and 4 give:

$$Q_{\text{th}} = \frac{2ALN}{60}$$

Result.

$$Q_{\text{th}} = \frac{ALN}{60} \ \text{(single-acting)}$$

$$Q_{\text{th}} = \frac{2ALN}{60} \ \text{(double-acting)}$$

What it tells us. The discharge is proportional to the piston area, the stroke and the speed. It does not contain the head, so a reciprocating pump delivers nearly the same flow at any delivery head.

If the piston-rod area a_r is taken into account for a double-acting pump, the rod side displaces less volume: $Q_{\text{th}} = (2A - a_r)\,LN/60$.

6.3 Work done and power CORE

IN SIMPLE WORDS

Work is force times distance. Here the pump lifts the weight of the liquid through the total height, suction head plus delivery head. Weight per second is $\rho g Q$. Multiply by the height to get the work per second, which is the power. Friction and acceleration are ignored for now. Section 7 adds them back.

Derivation from scratch: power of a reciprocating pump (no friction, no acceleration)

Aim. Find the power needed to lift the liquid through the static suction and delivery heads.

Assumptions.

1. Friction in the pipes and acceleration of the liquid are neglected.
2. There is no leakage, so the discharge equals Q_{th}.
3. The liquid is lifted through the static head $h_s + h_d$, where h_s is the height of the cylinder above the sump and h_d is the height of the delivery outlet above the cylinder.

Step 1. Weight of liquid lifted per second. The mass delivered per second is ρQ_{th}. Its weight is mass times g.

$$\text{Weight per second} = \rho g Q_{\text{th}}$$

Step 2. Work done per second. Work equals weight times the height through which it is lifted.

$$\text{Work per second} = \rho g Q_{\text{th}} \times \left(h_s + h_d \right)$$

Step 3. Substitute the discharge, single-acting. Use $Q_{\text{th}} = ALN/60$.

$$P = \rho g \times \frac{ALN}{60} \times \left(h_s + h_d \right) = \frac{\rho g ALN \left(h_s + h_d \right)}{60}$$

Step 4. Double-acting. Use $Q_{\text{th}} = 2ALN/60$. The discharge is doubled, so the power is doubled.

$$P = \frac{2\rho g ALN \left(h_s + h_d \right)}{60}$$

Result.

$$P = \frac{\rho g ALN \left(h_s + h_d \right)}{60} \text{ W (single-acting)}$$

$$P = \frac{2\rho g ALN \left(h_s + h_d \right)}{60} \text{ W (double-acting)}$$

What it tells us. The power rises in direct proportion to the discharge and to the total static head. Doubling the head at the same speed doubles the power but leaves the discharge unchanged.

6.4 Slip and coefficient of discharge CORE

IN SIMPLE WORDS

A real pump never delivers quite the full swept volume. Some liquid leaks past the piston and valves, and the valves close a little late. The lost flow is called slip. The coefficient of discharge C_d is simply the fraction of the theoretical flow that you actually get. A value near 0.95 is typical. Occasionally you get more than the theoretical flow, which is called negative slip.

The actual discharge Q_{act} is less than Q_{th} because of leakage past the piston and valves and delayed closing of the valves.

$$\text{Slip} = Q_{\text{th}} - Q_{\text{act}}, \qquad \%\ \text{slip} = \frac{Q_{\text{th}} - Q_{\text{act}}}{Q_{\text{th}}} \times 100 = \left(1 - C_d \right) \times 100, \qquad C_d = \frac{Q_{\text{act}}}{Q_{\text{th}}}$$

Negative slip. Sometimes $Q_{\text{act}} > Q_{\text{th}}$, giving negative slip ($C_d > 1$). This happens with a long suction pipe, a short delivery pipe and high speed: the momentum of the liquid column in the suction pipe keeps the delivery valve open and pushes liquid out before the delivery stroke begins.

Worked example

Step-by-step method.

1. Convert all lengths to metres and the speed to rpm. Find the piston area $A = \pi D^2/4$ in m^2.
2. Find the theoretical discharge from $Q_{\text{th}} = ALN/60$ (single-acting) or $2ALN/60$ (double-acting).
3. If the piston-rod area is given for a double-acting pump, use $(2A - a_r)\,LN/60$.
4. Find the slip as $Q_{\text{th}} - Q_{\text{act}}$, and $C_d = Q_{\text{act}}/Q_{\text{th}}$.
5. Find the power from $P = \rho g Q_{\text{th}}\,(h_s + h_d)$ with Q_{th} in m^3/s.
6. Convert the answer to the units asked (L/s, kW).

Example 6.0 (warm-up) CORE A single-acting pump has a piston of diameter 100 mm and a stroke of 200 mm. It runs at 60 rpm. The actual discharge is 1.5 L/s. Find the theoretical discharge and the coefficient of discharge.

1. $A = \dfrac{\pi}{4}(0.1)^2 = 0.007\,854\ \mathrm{m}^2$
2. $Q_{\text{th}} = \dfrac{ALN}{60} = \dfrac{0.007\,854 \times 0.2 \times 60}{60} = 0.001\,571\ \mathrm{m}^3/\mathrm{s} = 1.571\ \mathrm{L/s}$
3. $C_d = \dfrac{Q_{\text{act}}}{Q_{\text{th}}} = \dfrac{1.5}{1.571} = 0.955$

Example 6.1 CORE A single-acting reciprocating pump has a plunger diameter of 150 mm and stroke of 300 mm and runs at 50 rpm. The suction and delivery heads are 5 m and 20 m. The actual discharge is 4.0 L/s. Find the theoretical discharge, the slip, the coefficient of discharge and the theoretical power. What would the discharge and power be if the pump were double-acting?

1. $A = \dfrac{\pi}{4}(0.15)^2 = 0.017\,67\ \mathrm{m}^2$
2. $Q_{\text{th}} = \dfrac{ALN}{60} = \dfrac{0.017\,67 \times 0.3 \times 50}{60} = 0.004\,418\ \mathrm{m}^3/\mathrm{s} = 4.418\ \mathrm{L/s}$
3. $\text{Slip} = 4.418 - 4.0 = 0.418\ \mathrm{L/s}$; $\%\ \text{slip} = \dfrac{0.418}{4.418} \times 100 = 9.46\%$
4. $C_d = \dfrac{4.0}{4.418} = 0.905$
5. $P = \rho g Q_{\text{th}}\,(h_s + h_d) = 1000 \times 9.81 \times 0.004\,418 \times 25 = 1.08\ \mathrm{kW}$
6. Double-acting (rod area neglected): $Q_{\text{th}} = 8.84\ \mathrm{L/s}$, $P = 2.17\ \mathrm{kW}$

Common mistakes and exam tips.

- Using the piston diameter in mm in $A = \pi D^2/4$. Convert to metres first, otherwise the area is 10^6 times too large.
- Forgetting the factor 2 for a double-acting pump, or using $2A$ when the rod area is given. Use $(2A - a_r)$.
- Using Q in L/s in $P = \rho g Q H$. Convert to m^3/s first.
- Mixing up the stroke L and the crank radius r. The stroke is $L = 2r$.
- Taking slip as a percentage of the actual discharge. Slip is always measured against Q_{th}.

Check your understanding.

1. CORE A single-acting pump has $A = 0.01\ \mathrm{m}^2$, $L = 0.2\ \mathrm{m}$ and $N = 60$ rpm. What is Q_{th}? *(It is $0.01 \times 0.2 \times 60/60 = 0.002\ \mathrm{m}^3/\mathrm{s}$, or 2 L/s.)*
2. CORE If $C_d = 0.95$, what is the percentage slip? *(It is $(1 - 0.95) \times 100 = 5\%$.)*
3. HOTS The delivery head of a reciprocating pump is doubled at the same speed. What happens to the discharge and the power? *(The discharge stays nearly the same, since Q_{th} has no head term. The power rises, as it depends on $h_s + h_d$.)*

7. Indicator diagram, acceleration and friction (L7) CORE

The indicator diagram plots the pressure head in the cylinder against the piston's position over one revolution. Its area is proportional to the work done per cycle, so it shows how acceleration and pipe friction change the pressures and the power.

IN SIMPLE WORDS

The indicator diagram is a graph of cylinder pressure against piston position. Its area is the work done in one cycle, like the area under a pressure-volume graph for a gas. In the ideal case the graph is a plain rectangle. In a real pump the liquid in the pipes has to be speeded up and slowed down, and it rubs against the pipe wall. Both effects bend the rectangle. Acceleration tilts it. Friction adds curved bulges.

7.1 Ideal indicator diagram CORE

IN SIMPLE WORDS

With no acceleration and no friction the pressure in the cylinder stays constant during each stroke. During suction it sits below atmospheric pressure by the suction head. During delivery it sits above atmospheric pressure by the delivery head. Draw these two levels against the piston position and you get a rectangle. Its height is $h_s + h_d$ and its width is the stroke L.

With no acceleration and no friction, the cylinder pressure head is constant during each stroke:

- Suction stroke: $H_{\mathrm{atm}} - h_s$ (below atmospheric by the suction head)

- Delivery stroke: $H_{\text{atm}} + h_d$ (above atmospheric by the delivery head)

The diagram is a rectangle of height $(h_s + h_d)$ and width L:

$$\text{Work done per second} = \frac{\rho g A L N}{60}\ (\ h_s + h_d\ )\ \propto\ \text{area of indicator diagram}$$

7.2 Effect of acceleration of the liquid in the pipes CORE

IN SIMPLE WORDS

The piston does not move at a steady speed. It starts from rest, speeds up, then slows down to rest again. The long column of liquid in the pipe must do the same. Speeding up a heavy column needs an extra push, like pushing a loaded trolley from rest. That extra push is an extra pressure, and we write it as a head, h_a. The longer the pipe, the faster the crank and the bigger the piston compared with the pipe, the larger h_a becomes.

Derivation from scratch: acceleration head in the suction or delivery pipe

Aim. Find the extra pressure head h_a needed to accelerate the liquid column in a pipe of length L_p and area a, connected to a cylinder of piston area A.

Assumptions.

1. The piston moves with simple harmonic motion. This is true when the connecting rod is long compared with the crank radius r.
2. The crank turns at a constant angular speed $\omega = 2\pi N/60$.
3. The liquid is incompressible and fills the pipe, so every particle in the pipe moves with the same velocity at any instant.
4. The crank angle $\theta = \omega t$ is measured from the inner dead centre (the piston at the start of its suction stroke).

Step 1. Piston position. The piston follows the motion of the crank pin along the line of the stroke. At the inner dead centre the pin is a distance r from the crank centre. At angle θ its distance from the crank centre along that line is $r\cos\theta$. So the piston has moved a distance x from the inner dead centre:

$$x = r - r\cos\theta = r\ (\ 1 -\ \cos\theta\ )$$

Step 2. Piston velocity. Differentiate with respect to time. Use $\frac{d\theta}{dt} = \omega$ and $\frac{d}{d\theta}(\ \cos\theta\ ) = -\ \sin\theta$.

$$v_{piston} = \frac{dx}{dt} = \frac{dx}{d\theta}\frac{d\theta}{dt} = r\ \sin\theta \times \omega = \omega r\ \sin\theta$$

Step 3. Piston acceleration. Differentiate again, using $\frac{d}{d\theta}(\ \sin\theta\ ) = \cos\theta$ and ω constant.

$$a_{piston} = \frac{dv_{piston}}{dt} = \omega r\ \cos\theta \times \omega = \omega^2 r\ \cos\theta$$

Step 4. Continuity between piston and pipe. The volume the piston sweeps per second, $A\,v_{piston}$, must flow through the pipe of area a with velocity v_{pipe}.

$$A\,v_{piston} = a\,v_{pipe} \quad \Rightarrow \quad v_{pipe} = \frac{A}{a}\,v_{piston}$$

Step 5. Acceleration of the liquid in the pipe. Differentiate Step 4 with respect to time. The ratio A/a is constant.

$$a_{pipe} = \frac{A}{a}\,a_{piston} = \frac{A}{a}\,\omega^2 r\,\cos\theta$$

Step 6. Mass of the liquid column. The column has volume aL_p and density ρ.

$$m = \rho\,aL_p$$

Step 7. Force needed. By Newton's second law, force equals mass times acceleration.

$$F = m \times a_{pipe} = \rho\,aL_p \times \frac{A}{a}\,\omega^2 r\,\cos\theta$$

Step 8. Convert the force into a pressure head. This force acts over the pipe cross-section a, so the extra pressure is F/a. Dividing by ρg gives the head h_a.

$$h_a = \frac{F}{\rho g\,a} = \frac{\rho\,aL_p \times \dfrac{A}{a}\,\omega^2 r\,\cos\theta}{\rho g\,a} = \frac{L_p}{g}\,\frac{A}{a}\,\omega^2 r\,\cos\theta$$

Here ρ and one factor a cancel from the numerator and denominator.

Step 9. Maximum value. The factor $\cos\theta$ has its largest magnitude, 1, at $\theta = 0^\circ$ and $\theta = 180^\circ$. At these angles the piston is at the end of a stroke, where its acceleration is greatest.

Result.

$$x = r\,(\,1 - \cos\theta\,)\,, \qquad v_{piston} = \omega r\,\sin\theta, \qquad a_{piston} = \omega^2 r\,\cos\theta$$

$$h_a = \frac{L_p}{g}\,\frac{A}{a}\,\omega^2 r\,\cos\theta, \qquad h_{a,\,\max} = \frac{L_p}{g}\,\frac{A}{a}\,\omega^2 r \quad (\text{ at } \theta = 0^\circ \text{ and } 180^\circ\,)$$

What it tells us. The acceleration head grows with the pipe length, with the area ratio A/a and with the square of the speed. Doubling the speed makes it four times larger. It is zero at mid-stroke, where the piston moves at constant speed for an instant.

Using L_s, a_s for the suction pipe gives h_{as}; using L_d, a_d for the delivery pipe gives h_{ad}.

- **Suction stroke:** at the start ($\theta = 0$) the liquid must be accelerated, so the cylinder pressure is lowered by $h_{as,\,\max}$ to $H_{\text{atm}} - h_s - h_{as}$. At mid-stroke $h_a = 0$. At the end ($\theta = 180^\circ$) the liquid decelerates and the pressure rises to $H_{\text{atm}} - h_s + h_{as}$.
- **Delivery stroke:** at the start the pressure is raised to $H_{\text{atm}} + h_d + h_{ad}$; at the end it falls to $H_{\text{atm}} + h_d - h_{ad}$.

The rectangle becomes a parallelogram. Its area, and hence the work per cycle, is **unchanged**, because the extra work at the start of each stroke is returned at the end.

Derivation from scratch: acceleration does not change the work done per cycle

Aim. Show that the extra work done against the acceleration head over one stroke is zero.

Assumptions.

1. Simple harmonic motion, as in the derivation above.
2. On any stroke the acceleration head varies as $h_a = h_{a,\,\max}\,\cos\theta$, with the sign set by the direction of the acceleration.

Step 1. Force on the piston due to h_a. The extra pressure $\rho g h_a$ acts on the piston area A.

$$F_a = \rho g\, h_a\, A = \rho g\, A\, h_{a,\,\max}\,\cos\theta$$

Step 2. Small displacement of the piston. From $x = r\,(1 - \cos\theta)$ we get, on differentiating:

$$dx = r\,\sin\theta\, d\theta$$

Step 3. Extra work over the suction stroke. The suction stroke runs from $\theta = 0$ to $\theta = \pi$. Work is $\int F_a\, dx$.

$$W_a = \int_0^{\pi} \rho g\, A\, h_{a,\,\max}\,\cos\theta\;\, r\,\sin\theta\, d\theta = \rho g\, A\, r\, h_{a,\,\max} \int_0^{\pi} \sin\theta\,\cos\theta\, d\theta$$

Step 4. Evaluate the integral. Use $\int \sin\theta\,\cos\theta\, d\theta = \dfrac{\sin^2\theta}{2}$ and put in the limits.

$$\int_0^{\pi} \sin\theta\,\cos\theta\, d\theta = \left[\frac{\sin^2\theta}{2}\right]_0^{\pi} = \frac{\sin^2\pi}{2} - \frac{\sin^2 0}{2} = 0 - 0 = 0$$

Step 5. Delivery stroke. The same working with limits π to 2π gives $\left[\dfrac{\sin^2\theta}{2}\right]_{\pi}^{2\pi} = 0 - 0 = 0.$

Result.

$$W_a = \rho g\, A\, r\, h_{a,\,\max} \times 0 = 0 \quad \text{on each stroke}$$

What it tells us. In the first half of a stroke the piston has to push extra hard, and in the second half the liquid pushes back on the piston by the same amount. The two cancel exactly, so the area of the indicator diagram does not change. The peak force on the crank, however, does increase.

7.3 Effect of friction in the pipes CORE

IN SIMPLE WORDS

Moving liquid rubs against the pipe wall, and the rubbing wastes energy. Friction depends on the square of the speed. The piston is slow at the ends of the stroke and fast at the middle, so friction is zero at the ends and greatest at mid-stroke. Unlike acceleration, friction always works against the flow. The pump must supply this extra head on every stroke, so the work done goes up.

Derivation from scratch: friction head and power including friction

Aim. Find the friction head in a pipe during a stroke, and the power of a single-acting pump including the friction in the suction and delivery pipes.

Assumptions.

1. Simple harmonic motion of the piston, with $\theta = \omega t$ from the inner dead centre.
2. The friction head in a pipe is given by the Darcy formula $h_f = \dfrac{4 f L_p v^2}{2 g\, d}$, where f is the friction coefficient, L_p the pipe length, d its diameter and v the velocity in the pipe.
3. The velocity in the pipe is the instantaneous velocity from continuity, with no air vessel.

Step 1. Velocity in the pipe. From the previous derivation, $v_{pipe} = \dfrac{A}{a}\,\omega r \sin\theta$.

Step 2. Substitute in the Darcy formula.

$$h_f = \frac{4 f L_p}{2 g\, d}\left(\frac{A}{a}\,\omega r \sin\theta\right)^2$$

Step 3. Maximum value. The factor $\sin^2\theta$ is largest, equal to 1, at $\theta = 90^\circ$ (mid-stroke).

$$h_{f,\,\max} = \frac{4 f L_p}{2 g\, d}\left(\frac{A}{a}\,\omega r\right)^2, \qquad \text{so} \qquad h_f = h_{f,\,\max}\,\sin^2\theta$$

Step 4. Work done against friction on one stroke. The pump must push against the pressure $\rho g h_f$ over the area A through the displacement $dx = r\sin\theta\, d\theta$. The stroke runs from $\theta = 0$ to π.

$$W_f = \int_0^{\pi} \rho g\, A\, h_{f,\,\max}\,\sin^2\theta\; r\sin\theta\, d\theta = \rho g\, A\, r\, h_{f,\,\max}\int_0^{\pi}\sin^3\theta\, d\theta$$

Step 5. Evaluate $\int_0^{\pi}\sin^3\theta\, d\theta$. Write $\sin^3\theta = \sin\theta\left(1-\cos^2\theta\right)$ and integrate term by term.

$$\int_0^{\pi}\sin\theta\left(1-\cos^2\theta\right)d\theta = \left[-\cos\theta + \frac{\cos^3\theta}{3}\right]_0^{\pi}$$

$$= \left(1 - \frac{1}{3}\right) - \left(-1 + \frac{1}{3}\right) = \frac{4}{3}$$

Step 6. Mean friction head over the stroke. The mean head is the work divided by $\rho g\, A \times L$, with $L = 2r$.

$$\bar{h}_f = \frac{W_f}{\rho g\, A\ (\,2r\,)} = \frac{\rho g\, A\, r\, h_{f,\,\max} \times \frac{4}{3}}{2\rho g\, A\, r} = \frac{2}{3}\, h_{f,\,\max}$$

This is the same as saying that the mean ordinate of a parabola is two-thirds of its height.

Step 7. Power of the pump. The work per revolution of a single-acting pump is $\rho g A L$ times the mean total head. The suction stroke has the head $h_s + \frac{2}{3}h_{fs}$ and the delivery stroke has $h_d + \frac{2}{3}h_{fd}$. Multiply by $N/60$ revolutions per second.

$$P = \frac{\rho g A L N}{60}\left(\, h_s + \tfrac{2}{3}h_{fs} + h_d + \tfrac{2}{3}h_{fd}\,\right)$$

Result.

$$h_f = \frac{4 f L_p}{2g\, d}\left(\frac{A}{a}\,\omega r \sin\theta\right)^2, \qquad h_{f,\,\max} = \frac{4 f L_p}{2g\, d}\left(\frac{A}{a}\,\omega r\right)^2 \quad (\text{ at } \theta = 90^\circ\ )$$

$$P = \frac{\rho g A L N}{60}\left(\, h_s + h_d + \tfrac{2}{3}h_{fs} + \tfrac{2}{3}h_{fd}\,\right)\ \text{W (single-acting)}$$

What it tells us. Friction head goes as the square of the speed, so doubling the speed makes it four times larger. Because friction always opposes the flow, it adds to the work, unlike acceleration. Only two-thirds of the maximum friction head counts, because friction is small near the ends of the stroke.

Friction head depends on the square of the pipe velocity, $(\,A/a\,)\,\omega r \sin\theta$. It is zero at both ends of the stroke and largest at mid-stroke, so it adds a parabola to each line of the diagram: below the suction line and above the delivery line. Friction **increases** the work.

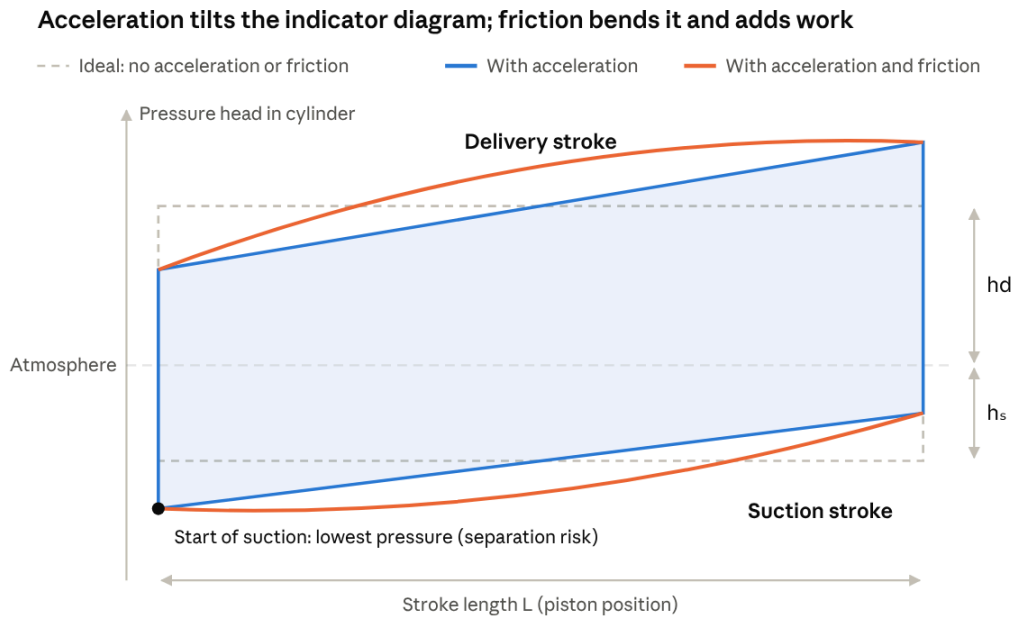


Figure 3. Indicator diagram of a single-acting reciprocating pump

The suction line slopes up because the cylinder pressure is lowest at the start of suction and highest at its end; the delivery line slopes the same way. Heights are not to scale.

7.4 Separation and the maximum speed HOTS

> **IN SIMPLE WORDS**
> If the pressure inside the cylinder drops too low, the liquid tears away from the piston and leaves a gap, like a vapour pocket. When the liquid comes back, it hits the piston with a bang and can damage the pump. The lowest pressure happens at the start of the suction stroke. To keep the pressure above the limit, we must keep the speed below a maximum. A faster pump or a longer suction pipe lowers the pressure more.

If the absolute pressure in the cylinder falls to the separation pressure head H_{sep} (about 2.5 m of water absolute), the liquid column separates from the piston. When it rejoins, it strikes the piston violently. The lowest pressures occur:

- At the **start of the suction stroke**: $H_{\text{atm}} - h_s - h_{as,\,\max}$ must be $\geq H_{\text{sep}}$
- At the **end of the delivery stroke**: $H_{\text{atm}} + h_d - h_{ad,\,\max}$ must be $\geq H_{\text{sep}}$

Both acceleration heads grow as ω^2, so each condition gives a maximum speed. The smaller of the two speeds governs.

Derivation from scratch: maximum speed without separation (suction side)

Aim. Find the largest crank speed $\omega_{\max}$ for which the cylinder pressure at the start of the suction stroke stays at or above H_{sep}.

Assumptions.

1. Simple harmonic motion, no air vessel on the suction side.
2. Friction in the suction pipe is neglected, which is the usual textbook simplification.
3. All heads are absolute heads of the liquid, in metres.

Step 1. Pressure at the start of the suction stroke. At $\theta = 0$ the cylinder pressure is the atmospheric head, minus the static suction lift, minus the acceleration head.

$$H_{cyl} = H_{\text{atm}} - h_s - h_{as,\ \max}$$

Step 2. Apply the no-separation condition. The pressure must not fall below H_{sep}.

$$H_{\text{atm}} - h_s - h_{as,\ \max} \geq H_{\text{sep}} \quad \Rightarrow \quad h_{as,\ \max} \leq H_{\text{atm}} - h_s - H_{\text{sep}}$$

Step 3. Insert the acceleration head. At the limiting speed, equality holds. Use $h_{as,\ \max} = \dfrac{L_s}{g}\dfrac{A}{a_s}\omega^2 r$ with the suction pipe data.

$$\frac{L_s}{g}\frac{A}{a_s}\omega_{\max}^2\, r = H_{\text{atm}} - h_s - H_{\text{sep}}$$

Step 4. Solve for $\omega_{\max}^2$. Divide both sides by $\dfrac{L_s}{g}\dfrac{A}{a_s}r$.

$$\omega_{\max}^2 = \frac{H_{\text{atm}} - h_s - H_{\text{sep}}}{\dfrac{L_s}{g}\dfrac{A}{a_s}r}$$

Step 5. Convert to rpm. Since $\omega = 2\pi N/60$, we have $N_{\max} = \dfrac{60\,\omega_{\max}}{2\pi}$.

Step 6. Delivery side (same method). At the end of the delivery stroke the condition is $H_{\text{atm}} + h_d - h_{ad,\ \max} \geq H_{\text{sep}}$, which gives $\omega_{\max}^2 = \dfrac{H_{\text{atm}} + h_d - H_{\text{sep}}}{\dfrac{L_d}{g}\dfrac{A}{a_d}r}$. The smaller of the two $\omega_{\max}$ values is the safe speed.

Result.

$$\omega_{\max}^2 = \frac{H_{\text{atm}} - h_s - H_{\text{sep}}}{\dfrac{L_s}{g}\dfrac{A}{a_s}r} \quad \text{(suction side)}$$

What it tells us. A higher suction lift, a longer suction pipe, a smaller pipe diameter or a longer stroke all reduce the safe speed. This is why long suction pipes need an air vessel near the pump.

Worked example

Step-by-step method.

1. Find $\omega = 2\pi N/60$, the crank radius $r = L/2$ and the area ratio $A/a = \left(D/d\right)^2$.
2. Find the acceleration head $h_{a,\,\max} = \left(L_p/g\right)\left(A/a\right)\omega^2 r$ for the suction pipe and for the delivery pipe.
3. Find the lowest cylinder pressure: $H_{\text{atm}} - h_s - h_{as,\,\max}$ at the start of suction, and $H_{\text{atm}} + h_d - h_{ad,\,\max}$ at the end of delivery.
4. Compare each with H_{sep}. If it is lower, the liquid separates.
5. For the maximum speed, set the lowest pressure equal to H_{sep}, solve for ω^2, then $N = 60\omega/2\pi$.
6. For friction, find $h_{f,\,\max}$ with the maximum pipe velocity and use $\frac{2}{3}h_{f,\,\max}$ in the power.

Example 7.0 (warm-up) CORE A single-acting pump has a piston of diameter 100 mm and a stroke of 200 mm, and runs at 60 rpm. The suction pipe is 50 mm in diameter and 2 m long. The suction lift is 3 m and $H_{\text{atm}} = 10.3\,\text{m}$. Find the maximum acceleration head in the suction pipe and the pressure head in the cylinder at the start of suction.

1. $\omega = \dfrac{2\pi \times 60}{60} = 6.283\,\text{rad/s}$; $r = 0.1\,\text{m}$; $\dfrac{A}{a_s} = \left(\dfrac{100}{50}\right)^2 = 4$
2. $h_{as,\,\max} = \dfrac{L_s}{g}\dfrac{A}{a_s}\omega^2 r = \dfrac{2}{9.81} \times 4 \times 6.283^2 \times 0.1 = 3.22\,\text{m}$
3. Pressure at start of suction $= 10.3 - 3 - 3.22 = 4.08\,\text{m}$ (absolute), which is above 2.5 m, so there is no separation

Example 7.1 HOTS The pump of Example 6.1 ($D = 150\,\text{mm}$, stroke 300 mm, 50 rpm, $h_s = 5\,\text{m}$, $h_d = 20\,\text{m}$) is fitted with suction and delivery pipes of 100 mm diameter, 4 m and 25 m long. Take $H_{\text{atm}} = 10.3\,\text{m}$ and $H_{\text{sep}} = 2.5\,\text{m}$. Check for separation and find the maximum safe speed.

1. $\omega = \dfrac{2\pi \times 50}{60} = 5.236\,\text{rad/s}$; $r = 0.15\,\text{m}$; $A/a = \left(150/100\right)^2 = 2.25$
2. $h_{as,\,\max} = \dfrac{4}{9.81} \times 2.25 \times 5.236^2 \times 0.15 = 3.77\,\text{m}$
3. Pressure at start of suction $= 10.3 - 5 - 3.77 = 1.53\,\text{m}$ (absolute), below 2.5 m: **separation occurs**
4. $h_{ad,\,\max} = \dfrac{25}{9.81} \times 2.25 \times 5.236^2 \times 0.15 = 23.58\,\text{m}$; pressure at end of delivery $= 10.3 + 20 - 23.58 = 6.72\,\text{m}$ (absolute), above 2.5 m: no separation on the delivery side
5. Suction limit: $h_{as,\,\max} = 10.3 - 5 - 2.5 = 2.8\,\text{m}$, so $\omega^2 = \dfrac{2.8}{\left(4/9.81\right) \times 2.25 \times 0.15} = 20.35$, $\omega = 4.511\,\text{rad/s}$, $N = 43.1\,\text{rpm}$
6. Delivery limit: $h_{ad,\,\max} = 10.3 + 20 - 2.5 = 27.8\,\text{m}$, so $\omega^2 = \dfrac{27.8}{\left(25/9.81\right) \times 2.25 \times 0.15} = 32.32$, $\omega = 5.685\,\text{rad/s}$, $N = 54.3\,\text{rpm}$

7. The suction side governs: $\text{maximum speed} = 43.1\ \text{rpm}$. To run at 50 rpm, fit a suction air vessel or use a larger, shorter suction pipe.

Common mistakes and exam tips.

- Using the diameter instead of the crank radius. Use $r = L/2$ in $h_a = \left(L_p/g \right) \left(A/a \right) \omega^2 r \cos\theta$.
- Forgetting to convert rpm to rad/s with $\omega = 2\pi N/60$. Also remember that h_a goes as ω^2.
- Taking A/a as a ratio of diameters. It is the ratio of areas, so square the diameter ratio.
- Using a negative absolute pressure as if it were real. A calculated value below zero only means that separation occurs.
- Saying that acceleration changes the work done. It does not; only friction does.

Check your understanding.

1. CORE At what crank angle is the friction head in a pipe largest? *(At $\theta = 90^\circ$, since $h_f \propto \sin^2\theta$.)*
2. CORE What is the acceleration head at mid-stroke? *(Zero, since $\cos 90^\circ = 0$.)*
3. HOTS The pump speed is doubled. By what factor do $h_{a,\ \max}$ and $h_{f,\ \max}$ change? *(Both become four times as large, since both depend on ω^2.)*

8. Air vessels (L8) CORE

An air vessel is a closed chamber with compressed air at the top and liquid at the bottom. It is fitted to the suction pipe close to the suction valve and to the delivery pipe close to the delivery valve. It acts as a cushion that smooths the flow in the pipes, so the long pipe columns move at nearly constant velocity.

IN SIMPLE WORDS

An air vessel works like the shock absorber of a car. The air inside is springy. It soaks up the surges when the piston pushes hard and gives liquid back when the piston is slow. The long pipe then sees a nearly steady flow instead of start-stop jerks. A steady flow means less friction and no violent acceleration of the long liquid column. The vessel therefore saves power and lets the pump run faster.

8.1 Functions CORE

IN SIMPLE WORDS

An air vessel does three jobs. It evens out the flow, it cuts the friction work, and it removes the acceleration head from the long pipes. The first job is easy to picture, since the pipe flow is smooth and not pulsing. The second follows because friction depends on velocity squared, and a steady velocity is much lower than the peak velocity. The third lets a pump run faster without the liquid separating.

1. Gives an almost **continuous, uniform flow** in the suction and delivery pipes
2. **Saves work** against friction: 84.8% of the friction work for a single-acting pump and 39.2% for a double-acting pump
3. Removes most of the **acceleration head** from the long pipes, so the pump can run at a **higher speed without separation**

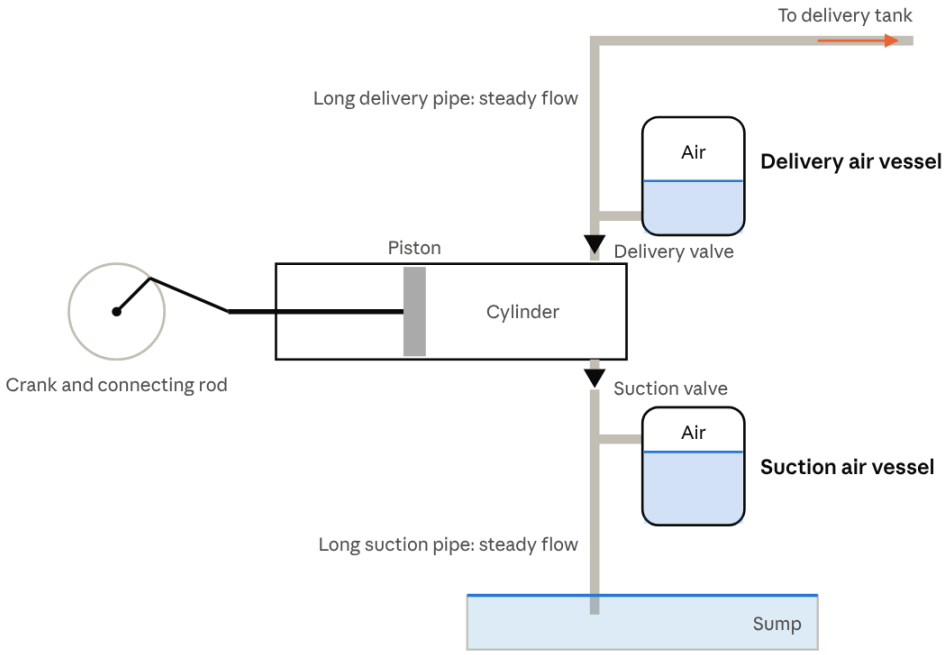
8.2 Working CORE

> **IN SIMPLE WORDS**
> Think of a water tank with a compressible cushion of air on top. When the piston delivers more than the average flow, the extra liquid enters the vessel and squeezes the air. When the piston delivers less than average, the squeezed air pushes liquid out to make up the difference. The air vessel on the suction side does the same job in reverse. Only the short piece of pipe between the vessel and the cylinder still sees the stop-start flow.

- **Delivery side.** During the first and last parts of the delivery stroke the piston moves slowly, so the piston delivers less than the mean flow and the air vessel supplies the difference. Around mid-stroke the piston moves fast and delivers more than the mean flow; the excess compresses the air in the vessel. During the suction stroke the vessel alone keeps the delivery pipe flowing.
- **Suction side.** The vessel stores liquid while the suction valve is closed and releases it into the cylinder during the suction stroke, so the long suction pipe flows steadily.

With air vessels, only the short lengths of pipe between the vessels and the cylinder carry the fluctuating flow.

Air vessels make the flow in the long pipes nearly steady



Only the short pipe lengths between the vessels and the cylinder carry the pulsating flow.

Figure 4. Single-acting reciprocating pump with suction and delivery air vessels

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

8.3 Rate of flow into and out of the air vessel ADVANCED

For a single-acting pump, the mean flow in the pipe is $ALN/60 = A\omega r/\pi$ (since $L = 2r$ and $\omega = 2\pi N/60$). The instantaneous piston delivery is $A\omega r \sin\theta$. So during the delivery stroke the flow into the delivery air vessel is:

$$q = A\omega r \sin\theta - \frac{A\omega r}{\pi} = A\omega r\left(\sin\theta - \frac{1}{\pi}\right)$$

Flow goes into the vessel when $\sin\theta > 1/\pi$ (crank angle between 18.6° and 161.4°) and out of it for the rest of the cycle. For a double-acting pump the mean flow is $2A\omega r/\pi$, giving $q = A\omega r\,(\sin\theta - 2/\pi)$.

8.4 Work saved against friction CORE

IN SIMPLE WORDS

Friction head depends on velocity squared. Without a vessel, the pipe velocity swings between zero and a peak, and the average friction is two-thirds of the peak value. With a vessel, the pipe velocity stays at the steady mean, which is much smaller than the peak. A small velocity squared is a very small friction head. For a single-acting pump the friction work falls by 84.8%. For a double-acting pump the fall is 39.2%.

Derivation from scratch: work saved against friction by an air vessel

Aim. Show that an air vessel saves 84.8% of the friction work in a single-acting pump and 39.2% in a double-acting pump.

Assumptions.

1. Simple harmonic motion of the piston, as in Section 7.
2. The friction head follows the Darcy formula $h_f = \dfrac{4fL_p v^2}{2g\,d}$.
3. With an air vessel placed close to the cylinder, the velocity in the long pipe is constant and equals the mean velocity. The short pipe between vessel and cylinder is neglected.

Step 1. Without an air vessel. The pipe velocity is $(A/a)\,\omega r \sin\theta$. Section 7.3 showed that the mean friction head over a stroke is two-thirds of the maximum.

$$\bar{h}_f = \frac{2}{3}h_{f,\,\max} = \frac{2}{3}\cdot\frac{4fL_p}{2g\,d}\left(\frac{A}{a}\omega r\right)^2$$

Step 2. Mean flow rate, single-acting. The pump delivers a volume AL in one revolution, which takes the time $T = 2\pi/\omega$. With $L = 2r$ the mean flow is:

$$Q_{\text{mean}} = \frac{AL}{T} = \frac{A \times 2r}{2\pi/\omega} = \frac{A\omega r}{\pi}$$

Step 3. Constant pipe velocity with the vessel. Divide the mean flow by the pipe area a.

$$v_{\text{mean}} = \frac{Q_{\text{mean}}}{a} = \frac{A}{a}\frac{\omega r}{\pi}$$

Step 4. Friction head with the vessel. Put v_{mean} in the Darcy formula and compare with $h_{f,\ \max}$.

$$h_f = \frac{4fL_p}{2g\,d}\left(\frac{A}{a}\frac{\omega r}{\pi}\right)^2 = \frac{1}{\pi^2}\frac{4fL_p}{2g\,d}\left(\frac{A}{a}\omega r\right)^2 = \frac{1}{\pi^2}\,h_{f,\ \max}$$

Step 5. Ratio of friction work, with to without. The friction work is proportional to the friction head, since the volume pumped is the same.

$$\frac{h_f\ (\text{ with })}{\bar{h}_f\ (\text{ without })} = \frac{1/\pi^2}{2/3} = \frac{3}{2\pi^2} = \frac{3}{19.739} = 0.152$$

Step 6. Saving, single-acting.

$$\text{Saving} = \ (\,1 - 0.152\,)\ \times 100 = 84.8\%$$

Step 7. Double-acting. Two delivery strokes per revolution double the mean flow, so $v_{\text{mean}} = \dfrac{2}{\pi}\dfrac{A}{a}\omega r$. The friction head with the vessel is $\dfrac{4}{\pi^2}\,h_{f,\ \max}$. The ratio is:

$$\frac{4/\pi^2}{2/3} = \frac{6}{\pi^2} = \frac{6}{9.870} = 0.608 \quad \Rightarrow \quad \text{Saving} = \ (\,1 - 0.608\,)\ \times 100 = 39.2\%$$

Result.

$$\text{Ratio with/without} = \frac{3}{2\pi^2} = 0.152 \Rightarrow 84.8\%\ \text{saved (single-acting)}$$

$$\frac{6}{\pi^2} = 0.608 \Rightarrow 39.2\%\ \text{saved (double-acting)}$$

What it tells us. The saving is larger for a single-acting pump because its flow in the pipe is more uneven, so the peak velocity is far above the mean. A double-acting pump has a smoother flow to begin with, so the vessel has less to improve.

Worked example

Step-by-step method.

1. Find $\omega = 2\pi N/60$, $r = L/2$ and $A/a = \left(D/d\right)^2$.
2. Find the maximum pipe velocity without a vessel: $v_{\max} = \left(A/a\right)\omega r$.
3. Find $h_{f,\ \max} = 4fL_p v_{\max}^2 / \left(2gd\right)$. The mean friction head without a vessel is $\frac{2}{3}h_{f,\ \max}$.
4. With a vessel, the velocity is $v_{\max}/\pi$ (single-acting) or $2v_{\max}/\pi$ (double-acting). Find h_f with this velocity.
5. Find the saving as $1 - h_f\left(\text{ with }\right) / \bar{h}_f\left(\text{ without }\right)$.
6. For the power saved, multiply the head saved by $\rho g Q_{\text{th}}$.

Example 8.0 (warm-up) CORE A single-acting pump has a piston of diameter 100 mm and a stroke of 200 mm, and runs at 60 rpm. The delivery pipe has a diameter of 50 mm. Find the maximum pipe velocity without an air vessel and the steady velocity with one.

1. $\omega = \dfrac{2\pi \times 60}{60} = 6.283\ \text{rad/s}$; $r = 0.1\ \text{m}$; $\dfrac{A}{a} = \left(\dfrac{100}{50}\right)^2 = 4$
2. $v_{\max} = \dfrac{A}{a}\omega r = 4 \times 6.283 \times 0.1 = \ 2.513\ \text{m/s}$
3. With the vessel (single-acting), $v_{\text{mean}} = \dfrac{v_{\max}}{\pi} = \dfrac{2.513}{\pi} = \ 0.8\ \text{m/s}$

Example 8.1 HOTS For the pump of Example 6.1 (50 rpm, $D = 150\ \text{mm}$, stroke 300 mm), the delivery pipe is 100 mm in diameter and 25 m long with $f = 0.01$. Find the friction head and the power spent on delivery-pipe friction with and without an air vessel.

1. Maximum pipe velocity $= \left(A/a\right)\omega r = 2.25 \times 5.236 \times 0.15 = 1.767\ \text{m/s}$
2. Without air vessel: $h_{fd,\ \max} = \dfrac{4 \times 0.01 \times 25 \times 1.767^2}{2 \times 9.81 \times 0.1} = 1.592\ \text{m}$; mean $= \frac{2}{3} \times 1.592 = \ 1.061\ \text{m}$
3. With air vessel: velocity $= 1.767/\pi = 0.5625\ \text{m/s}$; $h_{fd} = \dfrac{4 \times 0.01 \times 25 \times 0.5625^2}{2 \times 9.81 \times 0.1} = \ 0.161\ \text{m}$
4. Saving $= 1 - 0.161/1.061 = \ 84.8\%$, as predicted
5. Power saved $= \rho g Q_{\text{th}} \times \left(1.061 - 0.161\right) = 1000 \times 9.81 \times 0.004\,418 \times 0.900 = \ 39\ \text{W}$
6. With the air vessel, the acceleration head in the long delivery pipe also disappears, which removes the delivery-side speed limit found in Example 7.1

Common mistakes and exam tips.

- Quoting 84.8% for a double-acting pump. The saving is 84.8% for single-acting and 39.2% for double-acting.
- Using the maximum velocity in the “with air vessel” case. The steady velocity is $v_{\max}/\pi$, not $v_{\max}$.
- Forgetting the factor $\frac{2}{3}$ for the mean friction head without a vessel.
- Putting the vessel far from the cylinder. The saving is only valid when the vessel is close to the cylinder, because the pipe between them still carries the fluctuating flow.
- Mixing up the two sides. The suction vessel goes near the suction valve and the delivery vessel near the delivery valve.

Check your understanding.

1. CORE Name two functions of an air vessel. *(It gives uniform flow and saves friction work. It also removes the acceleration head from long pipes.)*
2. CORE What is the steady pipe velocity with an air vessel in a single-acting pump, in terms of the maximum velocity? *(It is $v_{\max}/\pi$.)*
3. HOTS Why is the saving smaller for a double-acting pump? *(Its flow is already smoother, so the mean velocity is closer to the peak: $2/\pi$ of the peak, not $1/\pi$.)*

9. Comparison of centrifugal and reciprocating pumps CORE

A centrifugal pump suits large, steady flows at moderate heads; a reciprocating pump suits small flows at high heads and is self-priming.

IN SIMPLE WORDS

A centrifugal pump is like a fast-spinning fan for water. It gives a big, smooth flow at modest pressure. A reciprocating pump is like a strong hand pump. It gives a small, pulsing flow, but it can reach very high pressure. The table below puts the two side by side. To choose, ask three questions. How much flow do you need? How high is the head? Does the liquid carry dirt?

Feature	Centrifugal pump	Reciprocating pump
Principle	Rotodynamic: kinetic energy given by impeller, converted to pressure	Positive displacement: piston pushes a fixed volume each stroke
Discharge	Large, continuous and uniform	Small, pulsating (smoothed by air vessels)
Head	Low to medium per stage; high only with multistage	High heads possible with one cylinder
Discharge–head relation	Discharge falls as head rises	Discharge almost independent of head
Speed	High, can be coupled directly to a motor	Low, needs gearing or belt drive
Priming	Required (unless self-priming or flooded)	Not required (self-priming)
Viscous liquids, liquids with solids	Can handle dirty water, sewage, slurry (open impeller)	Not suitable; valves get choked
Size, weight and cost for given discharge	Compact, light, cheaper	Bulky, heavy, costlier
Maintenance	Low, few wearing parts	Higher: valves, piston rings and glands wear
Efficiency	High at design point, falls away from it	High, nearly constant over wide range
Delivery valve closed while running	Safe for a short time: head rises to shut-off value	Dangerous: pressure builds until pipe or pump fails; needs a relief valve
Typical uses	Water supply, irrigation, process industries, boiler feed (multistage)	Hydraulic presses, small high-pressure duties, metering and dosing, oil-well pumping

Common mistakes and exam tips.

- Writing “reciprocating pumps give high discharge”. They give small discharge at high head. Centrifugal pumps give large discharge.
- Saying a centrifugal pump needs no priming. It must be primed, unless it is self-priming or the suction is flooded.
- Closing the delivery valve of a reciprocating pump. This is dangerous and needs a relief valve. For a centrifugal pump it is allowed briefly.
- Giving a comparison as a list of unrelated points. Write 6 to 8 points in a two-column table, one feature per row, for full marks.

Check your understanding.

1. CORE Which pump is self-priming? *(The reciprocating pump.)*
2. CORE Which pump gives a pulsating flow? *(The reciprocating pump.)*

3. HOTS Muddy water with sand must be pumped at a large flow and a low head. Which pump suits? *(A centrifugal pump with an open impeller, as valves in a reciprocating pump would choke.)*

10. Formula sheet and practice questions CORE

10.1 Formula sheet CORE

Topic	Result
Blade velocities	$u_1 = \pi D_1 N/60$, $u_2 = \pi D_2 N/60$
Outlet whirl	$V_{w2} = u_2 - V_{f2}/\tan\phi$
Discharge through impeller	$Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$
Work done per unit weight (radial entry)	$V_{w2} u_2 / g$
Manometric head	$H_m = h_s + h_d + h_{fs} + h_{fd} + V_d^2/(2g)$
Manometric efficiency	$\eta_{\text{mano}} = gH_m/(V_{w2}u_2)$
Overall efficiency	$\eta_o = \rho g Q H_m / P = \eta_{\text{mano}} \times \eta_m$
Minimum starting speed	$\left(u_2^2 - u_1^2\right)/(2g) = H_m$
Specific speed of pump	$N_s = N\sqrt{Q}\,/H_m^{3/4}$
Affinity laws (same pump)	$Q \propto N$, $H_m \propto N^2$, $P \propto N^3$
NPSH	$H_{\text{atm}} - H_v - h_s - h_{fs}$
Thoma's factor	$\sigma = \text{NPSH}/H_m$
Reciprocating discharge	$Q_{\text{th}} = ALN/60$ (single), $2ALN/60$ (double)
Slip	$\left(Q_{\text{th}} - Q_{\text{act}}\right)/Q_{\text{th}} \times 100$, $C_d = Q_{\text{act}}/Q_{\text{th}}$
Power (no friction)	$\rho g A L N (h_s + h_d)/60$
Acceleration head	$h_a = \left(L_p/g\right)(A/a)\,\omega^2 r\cos\theta$
Friction head	$h_f = \left(4fL_p/(2gd)\right)\left[(A/a)\,\omega r \sin\theta\right]^2$
Power with friction	$\rho g A L N (h_s + h_d + 2h_{fs}/3 + 2h_{fd}/3)/60$
Separation check	$H_{\text{atm}} - h_s - h_{a,\,\max} \geq H_{\text{sep}} \approx 2.5$ m (suction)
Air vessel saving	84.8% (single-acting), 39.2% (double-acting)

10.2 Short-answer questions (2 marks) CORE

1. CORE What is priming? Why is it necessary for a centrifugal pump?
2. CORE Define manometric head and manometric efficiency.
3. HOTS Why are the vanes of a centrifugal pump impeller curved backward?
4. CORE What is the function of the casing in a centrifugal pump? Name three types of casing.
5. CORE Define specific speed of a pump. How does it differ from that of a turbine?
6. CORE What is cavitation? State two of its effects.

7. CORE Define NPSH and Thoma's cavitation factor.
8. CORE What is slip in a reciprocating pump? When can it be negative?
9. HOTS Why is the work done unchanged by acceleration but increased by friction?
10. CORE State three functions of an air vessel.
11. HOTS Why should the delivery valve of a reciprocating pump never be closed while it runs?
12. CORE When are pumps connected in series and when in parallel?

10.3 Long-answer and numerical questions (10 marks) CORE

1. CORE With a neat sketch, explain the main parts and working of a centrifugal pump. Classify centrifugal pumps.
2. CORE Draw the inlet and outlet velocity triangles of a centrifugal pump impeller and derive the expression for work done per unit weight.
3. CORE Derive the expression for the minimum starting speed of a centrifugal pump.
4. CORE Explain cavitation in centrifugal pumps: causes, effects and methods of prevention. Define NPSH.
5. CORE Draw the ideal indicator diagram of a reciprocating pump and show the effect of acceleration and friction on it. Derive the expression for work done including friction.
6. CORE Explain the working of air vessels and prove that they save 84.8% of the work against friction in a single-acting pump.
7. HOTS Compare centrifugal and reciprocating pumps.
8. CORE A centrifugal pump impeller has diameters 200 mm and 400 mm and runs at 1200 rpm. The outlet vane angle is 30° and the velocity of flow is 2 m/s throughout, with radial entry. Find the inlet vane angle, work done per unit weight and the manometric head for $\eta_{\text{mano}} = 80\%$. *(Ans: 9.0°, $55.5\ \text{N}\cdot\text{m/N}$, 44.4 m)*
9. CORE A pump delivers $0.04\ \text{m}^3/\text{s}$ against a manometric head of 20 m with overall efficiency 75%. Find the shaft power. *(Ans: 10.5 kW)*
10. CORE Find the minimum starting speed of a pump with impeller diameters 300 mm and 600 mm working against 25 m. *(Ans: 814 rpm)*
11. CORE A pump runs at 1440 rpm, delivering $0.1\ \text{m}^3/\text{s}$ against 40 m. Find its specific speed and the type of impeller. *(Ans: 28.6, slow radial flow)*
12. HOTS A pump works against $H_m = 50\ \text{m}$ with suction lift 3.5 m and suction friction 1.0 m. $H_{\text{atm}} = 10.3\ \text{m}$, $H_v = 0.3\ \text{m}$, $\sigma_c = 0.12$. Check for cavitation and find the maximum suction lift. *(Ans: $\sigma = 0.11 < 0.12$, cavitation; $h_{s,\ \max} = 3.0\ \text{m}$)*
13. CORE A double-acting pump has piston diameter 200 mm, piston rod diameter 40 mm, stroke 400 mm and speed 40 rpm. Actual discharge is 15.5 L/s. Find the theoretical discharge, % slip and C_d. *(Ans: 16.42 L/s, 5.6%, 0.944)*
14. HOTS A single-acting pump has plunger diameter 100 mm and stroke 200 mm. The suction pipe is 75 mm in diameter and 6 m long, and the suction lift is 4 m. Find the maximum speed without separation ($H_{\text{atm}} = 10.3\ \text{m}$, $H_{\text{sep}} = 2.5\ \text{m}$). *(Ans: 56.5 rpm)*